

1 Embedded Artificial Intelligence in Battery Management
2 Systems: Pruning and Quantization for Efficient
3 State-of-Charge and State-of-Health Estimation

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6 **Abstract**

7 Online battery management requires accurate state-of-charge (SOC) and state-
8 of-health (SOH) estimates within strict memory and timing budgets. This study
9 investigates an engineering application of artificial intelligence (AI): continu-
10 ous battery-state estimation on an STM32H753ZI microcontroller. The im-
11 plemented AI consists of stateful long short-term memory (LSTM) networks
12 with multi-layer perceptron (MLP) output heads. Using an open lithium iron
13 phosphate (LFP) ageing dataset, we compare full-precision Base models with
14 structured-pruned and recurrent-weight-quantized variants under identical long-
15 horizon replay and hardware measurements. This paired, auditable compression
16 benchmark is the AI contribution of the work. SOC mean absolute error (MAE)
17 ranges from 2.3 to 2.8 percentage points, while SOH MAE ranges from 0.9 to 1.5
18 percentage points. Pruning reduces flash by 40.9 % for SOC and 45.5 % for SOH
19 and reduces inference time by 42.9 % and 44.0 %, respectively. Quantization
20 reduces flash by 50.2 % and 58.8 %, but the mixed-precision kernel increases
21 inference time. Estimated energy is a constant-power timing proxy, not a sepa-
22 rate power measurement. The quantitative conclusions apply to the evaluated
23 LFP data, model configurations, firmware, and build settings. Additional cell
24 chemistries and embedded platforms require separate validation.

25 *Keywords:* Embedded artificial intelligence, Battery state estimation, Long
26 short-term memory network, Structured pruning, Weight quantization,
27 Microcontroller deployment

28 **Nomenclature**

29 *Acronyms*

30	BMS	Battery management system
31	SOC	State of charge
32	SOH	State of health
33	DoE	Design of Experiments
34	LFP	Lithium iron phosphate
35	LSTM	Long short-term memory network
36	MLP	Multi-layer perceptron
37	STM32	32-bit ARM Cortex-M microcontroller family
38	MAE	Mean absolute error
39	RMSE	Root-mean-square error
40	MAC	Multiply–accumulate operation
41	INT8	Signed 8-bit integer representation (quantized weights)
42	FP32	32-bit floating-point representation

43 *Symbols*

44	t	Time
45	$U(t)$	Terminal voltage at time t
46	$I(t)$	Current at time t (positive for discharge)
47	$T(t)$	Cell temperature at time t
48	$\text{EFC}(t)$	Equivalent full cycles accumulated up to t
49	$Q_c(t)$	Cumulative charge throughput up to t
50	C_{nom}	Nominal discharge capacity of the cell
51	C_{app}	Apparent capacity inferred from the latest check-up
52	DoD	Depth of discharge

53	$\text{SOC}(t)$	State of charge at time t
54	SOH_k	State of health at reference cycle k
55	$\mathbf{x}_t$	Input feature vector at time t
56	$\mathbf{h}_t$	LSTM hidden state at time t
57	$\mathbf{c}_t$	LSTM cell state at time t
58	$\mathbf{z}_t$	Concatenation of input and previous hidden state at time t
59	$\mathbf{i}_t, \mathbf{f}_t, \mathbf{o}_t, \mathbf{g}_t$	Input, forget, output, and candidate gates at time t
60	$\sigma(\cdot)$	Logistic sigmoid activation function
61	$\tanh(\cdot)$	Hyperbolic tangent activation function
62	$\odot$	Element-wise (Hadamard) product
63	W	Weight matrix (LSTM or MLP)
64	b	Bias vector
65	q	Quantized weight (INT8 code)
66	s	Per-row quantization scale (FP32)
67	$\hat{w}$	Reconstructed floating-point weight after dequantization
68	ε_q	Positive lower bound for the quantization scale
69	$\mathcal{R}$	Index set of retained LSTM hidden units
70	e_n	Signed prediction error of sample n in percentage points
71	$Q_p(\cdot)$	Empirical quantile operator at probability p
72	$W_{\text{ih}}^{\text{int8}}$	Quantized input-to-gate weight matrix
73	$W_{\text{hh}}^{\text{int8}}$	Quantized hidden-to-gate weight matrix
74	$s_{\text{ih}}, s_{\text{hh}}$	Per-row quantization scale vectors for $W_{\text{ih}}^{\text{int8}}$ and $W_{\text{hh}}^{\text{int8}}$
75	b_{fp32}	Floating-point bias of a given gate
76	g	Gate pre-activation used in the quantized LSTM kernel
77	D, H, M	Input dimension, LSTM hidden size, and MLP width
78	N_{MAC}	Analytical multiply-accumulate count per inference

79 1. Introduction

80 Accurate online estimation of the state of charge (SOC) and the state of
81 health (SOH) remains a central challenge for battery-powered systems [1, 2, 3, 4].
82 Recent reviews show sustained progress in model-based and data-driven estima-
83 tors, but they also emphasise that accuracy depends strongly on cell chemistry,
84 operating conditions, reference construction, and evaluation protocol [5, 1]. For
85 an embedded battery management system, predictive accuracy alone is there-
86 fore insufficient: the estimator must also fit the available flash and RAM, meet
87 the update deadline, and preserve its state consistently over long input streams.

88 The study is motivated by modular stationary storage systems in remote
89 microgrids. Within the MG Farm programme [6, 7, 8, 9], partners from France,
90 Algeria, Morocco, and Germany develop smart-grid demonstrators for North
91 African sites. The Technical University of Berlin contributes the battery ageing
92 experiments, the LFP DoE Cycle Ageing Dataset, a battery management sys-
93 tem prototype, and the SOC/SOH estimators deployed on its controller. Each
94 storage module must operate autonomously within the flash, RAM, and timing
95 limits of an STM32H753ZI-class microcontroller. Similar constraints arise in
96 mobile systems that cannot rely on continuous cloud access.

97 Existing embedded studies demonstrate that compact neural SOC or SOH
98 estimators are feasible on microcontrollers and that quantization can reduce
99 their storage requirements [10, 11, 12, 13, 14]. However, these studies generally
100 examine one battery state, one model family, or one deployment platform. They
101 do not provide a common, auditable comparison of structured hidden-unit prun-
102 ing and recurrent-weight quantization for both stateful SOC and SOH estima-
103 tors under identical long-horizon replay and hardware measurement conditions.
104 Reproduction is further hindered when scaling parameters, recurrent-state han-
105 dling, and the exact mixed-precision boundary are not reported.

106 This work uses the Lithium Iron Phosphate (LFP) DoE Cycle Ageing Dataset [15].
107 It contains long-term cyclic ageing measurements from multiple LFP cells under
108 full and fractional factorial conditions, with controlled temperature, charge and

109 discharge rates, depth-of-discharge variations, and interleaved capacity, open-
 110 circuit-voltage, and pulse-power check-ups. These data support a paired assess-
 111 ment of rapid SOC dynamics and slowly evolving SOH trajectories.

112 The objective of this study is to quantify, under identical data and deploy-
 113 ment conditions, how structured hidden-unit pruning and recurrent-weight post-
 114 training quantization change the accuracy, memory footprint, and inference cost
 115 of stateful LSTM + MLP SOC and SOH estimators on an STM32H753ZI. The
 116 study makes three contributions:

- 117 (i) an auditable path from trained PyTorch models to stateful C inference,
 118 including explicit feature scaling, recurrent-state handling, pruning, and
 119 mixed-precision quantization.
- 120 (ii) a controlled Base–Pruned–Quantized comparison for both SOC and SOH
 121 using continuous replay, linked-firmware flash and RAM, host latency, and
 122 cycle-counter-measured kernel time.
- 123 (iii) an evidence-bounded trade-off analysis that connects architecture-level
 124 operation counts and implementation details to long-horizon accuracy,
 125 filtering, and application-dependent utility.

126 The novelty lies in this paired, transparent deployment and compression bench-
 127 mark rather than in proposing a new recurrent cell or compression algorithm.

128 2. Related Work and Motivation

129 2.1. SOC/SOH Estimation

130 Existing SOC/SOH estimation methods can be grouped into three fami-
 131 lies [2, 3, 4, 5]. First, *measurement-derived* techniques use Coulomb counting
 132 and open-circuit-voltage (OCV) look-up tables during rest periods. They pro-
 133 vide inexpensive proxies but suffer from drift, offset accumulation, and a strong
 134 dependence on initialisation and periodic recalibration. Second, *model-based*
 135 observers combine equivalent-circuit or electrochemical models with Kalman fil-
 136 tering and its variants. These approaches can offer physical interpretability but
 137 require careful parameter identification and are sensitive to model mismatch and

138 ageing-induced parameter drift. Third, *data-driven* methods employ machine-
 139 learning regressors, including LSTM networks, Transformers, and hybrid struc-
 140 tures, to map sequences of current, voltage, and temperature to SOC and SOH
 141 estimates. Key challenges for such models are robustness when operated un-
 142 der conditions that differ from the training data and the careful treatment of
 143 normalisation and state handling on embedded targets.

144 Recent hybrid estimators combine physical or statistical preprocessing with
 145 learned correction models. Wang et al. [16] couple online parameter identifi-
 146 cation with an adaptive radial-basis correction and differential support-vector
 147 machine for SOC estimation under dynamic vehicle-oriented tests. For whole-
 148 life capacity estimation, Wang et al. [17] combine singular filtering, Gaussian-
 149 process regression, and an LSTM to address fast ageing and varying currents.
 150 These studies broaden the operating conditions and model structures consid-
 151 ered for battery-state estimation. Their primary evidence concerns estimation
 152 accuracy, however. Neither reports a paired post-training pruning and quantiza-
 153 tion study with linked-firmware memory and recurrent microcontroller timing.
 154 They are therefore complementary to, rather than substitutes for, the deploy-
 155 ment benchmark developed here. Across these families the reported accuracy
 156 spans a wide range. Electrochemical or Equivalent Circuit Model (ECM)-based
 157 observers with carefully tuned filters often achieve SOC root-mean-square errors
 158 (RMSE) well below 1 % of full scale and sub-percent SOH capacity errors on lab-
 159 oratory test cycles [18, 19, 20, 21]. Recent sequence-model approaches based on
 160 N-BEATS and Bidirectional LSTMs (BiLSTMs) for sodium-ion cells report SOC
 161 Mean Absolute Error (MAE) between roughly 0.1 and 0.4 % [22, 23], whereas
 162 TinyML-style SOC estimators deployed on microcontrollers often operate in the
 163 1–3 % error regime [10, 11, 12]. These values provide context rather than a di-
 164 rect ranking because chemistry, data splits, target construction, drive cycles,
 165 and error definitions differ between studies. Kalman filters, particle filters, and
 166 equivalent-circuit-model observers therefore dominate the classical literature,
 167 yet their accuracy can degrade over long operating periods with pronounced
 168 current and temperature gradients [19, 21]. Recurrent neural networks, and in

particular LSTM-based regressors, address these limitations by integrating long
 temporal context and absorbing residual modelling errors [24, 21]. In this work
 we deliberately restrict the architecture search to a single-layer LSTM with a
 fully connected Multi-Layer Perceptron (MLP) output layer. LSTMs retain ex-
 plicit cell states with separate input, forget, and output gates, which made it
 straightforward to hand-code a numerically stable, stateful kernel in C and to
 reuse the same backbone for both SOC and SOH. Gated recurrent units (GRUs)
 or Transformer variants could offer alternative trade-offs between accuracy and
 cost, but exploring multiple architectures would dilute the focus on the embed-
 ded tool flow and is left as future work. Most publications on LSTM-based
 estimators maximise accuracy with complex architectures and rarely examine
 how the resulting models behave once deployed on constrained controllers. Even
 within the LSTM family, sliding-window inference is commonly employed be-
 cause it delivers high accuracy, but it also requires substantial RAM to buffer
 multiple past samples. A stateful LSTM that hands over its hidden and cell
 vectors step by step retains the temporal information without maintaining large
 windows and therefore forms the first optimisation step in our architecture.
 The associated handling of hidden states, feature scaling, and numerical stabil-
 ity is rarely described in the literature, which is why this work explicitly targets
 embedded SOC/SOH deployment.

2.2. Embedded Deployment Strategies

Commercial toolchains such as ONNX Runtime, TensorRT, and STM32Cube.AI
 support many neural-network deployment workflows [25, 26, 27]. For continu-
 ously stateful estimators, however, a generated binary does not always expose
 recurrent-state handling or the precise placement of mixed-precision scaling op-
 erations needed for an implementation audit. Naguib et al. [14] compare accu-
 racy, execution time, flash, and RAM for classical and neural SOC estimators
 on two NXP microcontrollers. Crocioni et al. [10] deploy compact recurrent ca-
 pacity estimators on STM32 targets and study 8-bit quantization. More recent
 TinyML studies evaluate quantized or hardware-accelerated SOC estimators

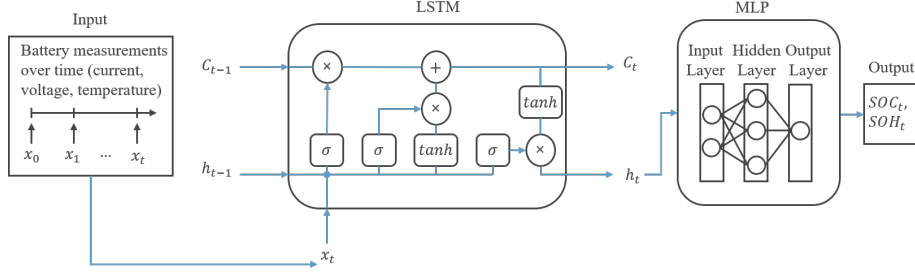


Figure 1: Schematic of the deployed SOC/SOH estimator. Preprocessed feature streams are fed into a single-layer LSTM whose hidden state is propagated across time. At each step a lightweight MLP head maps the hidden representation either to SOC or SOH, enabling stateful streaming inference on the STM32.

on embedded platforms [11, 12], while Kamali et al. [13] target computation-
ally compact SOH estimation. These studies establish embedded feasibility,
but each addresses a narrower combination of task, architecture, compression
method, and target hardware.

At the compression-method level, PQ Bench [28] systematically compares
pruning and quantization across general deep-learning workloads and confirms
that their accuracy-storage trade-offs are model dependent. It does not ad-
dress the continuous recurrent state, SOH post-processing, or microcontroller
timing behaviour of battery estimators. The resulting research gap is therefore
not the absence of compact neural estimators, but the absence of a controlled,
task-paired benchmark that traces two transparent compression routes from
trained SOC and SOH models to measured stateful firmware behaviour. Our
study addresses this gap with the same input dimension, streaming protocol,
performance-indicator definitions, and STM32 measurement procedure for Base,
Pruned, and Quantized variants of both tasks.

To retain control over the implementation boundary, we extract the Py-
Torch tensors, remove complete recurrent channels according to the documented
saliency score, apply per-row symmetric quantization to the recurrent weights,
and implement the LSTM gates and activations in C. This path makes the state

218 update, scaling operations, and stored parameter types inspectable and links the
 219 model transformations directly to the reported flash, RAM, and timing mea-
 220 surements.

221 **3. Dataset and Preprocessing**

222 *3.1. Ageing Experiments and Load Profiles*

223 All experiments rely on the high-resolution LFP DoE Cycle Ageing Dataset [15].
 224 The measurements originate from a design-of-experiments study on commer-
 225 cially available JGCFR18650-1800mAh-3.2V LFP cells [29] conducted at the
 226 Technical University of Berlin. Cells were cycled inside a climatic chamber at
 227 25°C while the following factors were systematically varied:

$$C_{\text{chg}} \in [0.5 \text{ C}, 0.9 \text{ C}], \quad (1)$$

$$C_{\text{dis}} \in [1.0 \text{ C}, 2.5 \text{ C}], \quad (2)$$

$$\text{DoD} \in [45\%, 65\%]. \quad (3)$$

228 The mixed full/fractional factorial plan distributes fifteen test cells across eight
 229 operating-point combinations and includes replication to assess repeatability.
 230 Figure 2 shows the low and high levels of all three factors.

231 Between ageing loops, dedicated check-ups provide capacity tests, open-
 232 circuit-voltage sweeps, and hybrid pulse-power characterisation. Stationary-
 233 storage profiles are additionally injected to emulate realistic power flows [30].
 234 These profiles originate from residential photovoltaic home-storage applications
 235 and contain stochastic load and generation patterns that excite the cell differ-
 236 ently from constant-current cycling.

237 The raw dataset contains uniformly sampled measurements at 1 Hz, includ-
 238 ing voltage, current, terminal temperature, and auxiliary channels. Partitioned
 239 Parquet tables support efficient retrieval of the long ageing sequences. The
 240 dataset is published in [15]. The cube notation follows standard DoE conven-
 241 tions as summarised in [31]. Together, Figures 3 and 4 illustrate the diversity of

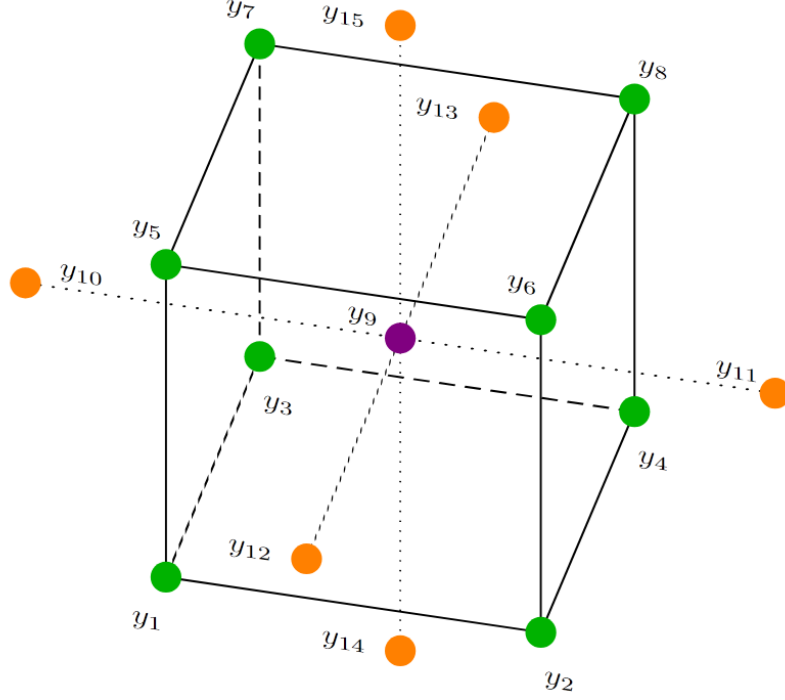


Figure 2: General three-factor design-of-experiments geometry used to structure the LFP ageing campaign. The cube represents the factorial design space, while the generic labels y_1 to y_{15} denote experimental conditions within and around it. The dataset-specific charge rate, discharge rate, and depth-of-discharge levels are described in the accompanying text.

242 SOC transients and ageing behaviour that the estimators must capture: short-
 243 term SOC dynamics are driven by fast current swings and mode changes between
 244 profiles, whereas SOH evolves slowly over hundreds of days and depends on the
 245 cumulative operating conditions of each cell.

246 3.2. Feature Engineering

247 The SOC model consumes the six-dimensional feature vector

$$\mathbf{x}_t^{\text{SOC}} = [U(t), I(t), T(t), Q_c(t), \frac{dU}{dt}(t), \frac{dI}{dt}(t)], \quad (4)$$

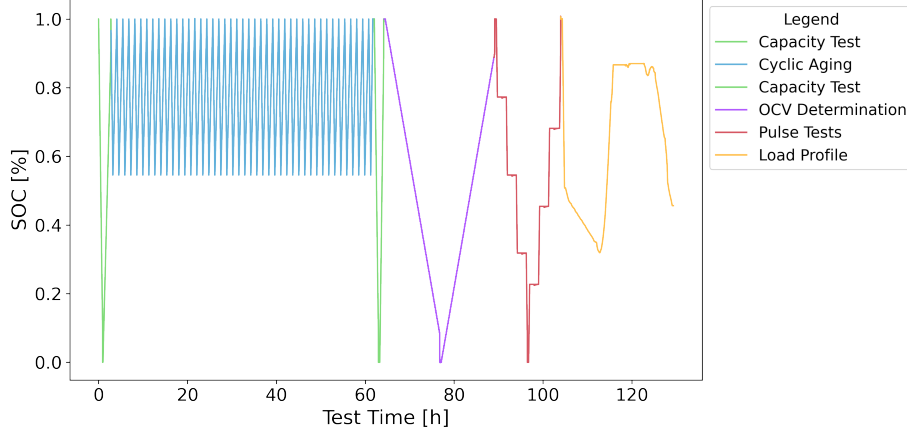


Figure 3: Representative SOC trajectory illustrating selected segments of the experimental protocol used in this study. The plot shows a portion of the cyclic ageing procedure with defined charge and discharge currents and a specified depth-of-discharge (DoD), followed by the diagnostic checkup sequence comprising OCV determination, HPPC pulses, and capacity tests. In addition, the trajectory includes a load profile taken from a stationary battery storage system. Together, these segments capture both controlled laboratory diagnostics and application-oriented operating conditions.

248 where the audited preprocessing uses timestamp-aware backward differences
 249 rather than centred differences. For sample index $k > 0$,

$$\Delta t_k = \max(t_k - t_{k-1}, \varepsilon_t), \quad \varepsilon_t = 10^{-6}, \quad (5)$$

$$\left. \frac{dU}{dt} \right|_k = \frac{U_k - U_{k-1}}{\Delta t_k}, \quad \left. \frac{dI}{dt} \right|_k = \frac{I_k - I_{k-1}}{\Delta t_k}. \quad (6)$$

250 This causal calculation requires only the preceding voltage/current sample and
 251 timestamp and has constant work per update. In the evaluated benchmark, how-
 252 ever, derivative and cumulative channels were prepared host-side before replay.
 253 Complete six-feature vectors were transmitted via UART, and the firmware
 254 applied the same robust-scaling parameters used during model development.
 255 Consequently, the reported DWT kernel times exclude sensor acquisition and
 256 feature construction. The exact placement of the six-feature scaling operation
 257 is documented with the STM32 measurement interval below. Each component
 258 is normalised using a robust scaling operation based on the median m and in-

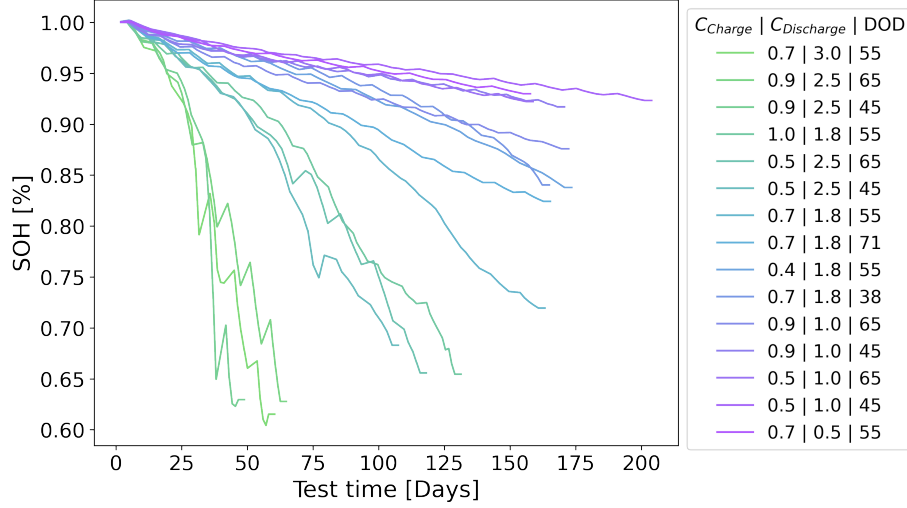


Figure 4: Evolution of the state of health across all LFP cells in the ageing campaign, shown over the total experimental duration. The spread between trajectories reflects the different operating points in the factorial plan (C-rates, depth of discharge, profile mix) and motivates the use of a data-driven SOH estimator that generalises across cells and loading conditions.

terquartile range r :

$$\tilde{x}_{t,k}^{SOC} = \frac{x_{t,k}^{SOC} - m_k}{r_k + \varepsilon}, \quad \varepsilon = 10^{-6}. \quad (7)$$

The same (m_k, r_k) pairs are embedded in the firmware so that host and STM32 inputs are normalised identically. The SOH model uses a slightly different feature stack to capture slow-varying degradation markers:

$$\mathbf{x}_t^{SOH} = [t, U(t), I(t), T(t), EFC(t), Q_c(t)], \quad (8)$$

where t is the cumulative time stamp and $EFC(t)$ denotes the equivalent full cycles accrued up to sample t . These channels are likewise robustly normalised, relying on median and interquartile range rather than mean and variance to reduce the influence of outliers in current and voltage measurements [32, 33]. During deployment, the SOC and SOH models run independently on the microcontroller and each produces one estimate per second. In principle, both estimates could be fused more tightly, but this work keeps the models separate to evaluate their resource use and accuracy independently.

271 3.3. Reference SOC and SOH targets

272 The ground-truth SOC trajectory is derived from Coulomb counting with
 273 drift compensation. Denoting the apparent cell capacity after a given check-up
 274 by C_{app} , the SOC inside a charge-discharge segment is computed as

$$\text{SOC}(t) = \text{SOC}(t_0) - \frac{1}{C_{\text{app}}} \int_{t_0}^t I(\tau) \text{d}\tau, \quad (9)$$

275 where the integral is reset to $\text{SOC} = 1$ after each constant-voltage (CV) ter-
 276 mination. The reset mitigates the bias that would otherwise accumulate due
 277 to capacity fade, yet the long-term traces still reveal gradual changes in the
 278 SOC envelope as C_{app} decreases. State of health is obtained from the periodic
 279 capacity check-ups through

$$\text{SOH}_k = \frac{C_k}{C_{\text{nom}}}, \quad (10)$$

280 with C_k representing the measured discharge capacity during the k -th reference
 281 cycle (comprising capacity test, open-circuit-voltage sweep, and hybrid pulse-
 282 power characterisation) relative to the nominal capacity C_{nom} . These targets
 283 highlight the need for a data-driven estimator that remains accurate despite
 284 the evolving capacity landscape, which motivates the use of recurrent neural
 285 networks for both SOC and SOH predictions. The reference SOC and SOH
 286 traces are therefore offline constructs. They rely on information that is not
 287 available in deployment, such as long rest periods for OCV stabilisation and
 288 capacity from periodic full cycles, and they assume perfect initialisation and
 289 recalibration. These constructions follow common practice in battery manage-
 290 ment, where SOC is obtained by Coulomb counting and SOH is expressed as the
 291 ratio between present and nominal capacity [34, 2, 3]. On the STM32 the esti-
 292 mators operate purely on the streamed measurements and their internal states.
 293 The reference trajectories only serve as ground truth for training and evaluation.

294 3.4. Streaming inference setup

295 Because the SOC and SOH estimators are stateful, the LSTM hidden and
 296 cell states are continuously forwarded from one sample to the next along the full

input stream. At each sampling instant the current feature vector is processed together with the previous hidden and cell states, which yields the new states and the corresponding SOC or SOH estimate. We therefore operate the models in a purely step-by-step streaming regime without forming sliding input windows. On the host the full normalised sequence from the LFP DoE Cycle Ageing Dataset is processed once in this fashion, and the resulting SOC and SOH estimates are aligned with the reference trajectories. On the STM32 firmware the same step-wise LSTM implementation is used, driven by the UART feature stream, so that host and embedded benchmarks reflect identical inference conditions. The streaming dashboards in Figs. 8 and 11 illustrate a complete run for one representative cell.

3.5. SOH prediction post-processing

The SOH trajectories in the final dashboard use two sequential causal filtering stages. First, each raw model output $\hat{y}_t$ is multiplicatively aligned with the initial SOH label, $\hat{y}_t^{(0)} = (y_0/\hat{y}_0)\hat{y}_t$. Stage 1 limits the change of this calibrated prediction relative to the preceding stage output to

$$\ell_t = \min\left(10^{-4}|y_{t-1}^{(1)}|, 10^{-5}\right), \quad (11)$$

and subsequently applies an exponential moving average (EMA) with $\alpha_1 = 0.02$. The limiter is symmetric, so positive and negative proposal changes are both restricted to $\pm\ell_t$. Stage 2 consumes the complete stage-1 trajectory, applies another EMA with $\alpha_2 = 10^{-6}$, and limits only downward changes according to

$$y_t^{(2)} = \max\left(e_t^{(2)}, y_{t-1}^{(2)} - 2 \times 10^{-8}\right), \quad e_t^{(2)} = (1 - \alpha_2)e_{t-1}^{(2)} + \alpha_2 y_t^{(1)}. \quad (12)$$

Thus, the curve in Fig. 11 is not obtained by applying α_2 directly to an unfiltered network output.

For an EMA sampled at interval Δt , the equivalent time constant and cutoff frequency are

$$\tau = -\frac{\Delta t}{\ln(1 - \alpha)}, \quad f_c = \frac{1}{2\pi\tau}. \quad (13)$$

At the dataset rate of 1 Hz, stage 1 has $\tau = 49.5$ s, $f_c = 3.22$ mHz, and an EMA-only 90 % step-response time of approximately 114 s. For stage 2, τ is approximately 11.57 days, $f_c = 1.59 \times 10^{-7}$ Hz, and the corresponding 90 % time is 26.65 days. These frequencies characterise only the linear EMA components. The nonlinear rate limiters have no single cutoff frequency. The very strong second stage can reduce error against slowly varying labels while introducing substantial response delay, which is evaluated separately in Appendix A.3.

4. Model Architecture and Baseline Performance

4.1. LSTM architecture in PyTorch

Both SOC and SOH estimators employ the same recurrent equation set but differ in size according to their training releases. The SOC configuration uses $H = 64$ hidden units and an MLP output layer of width 64, whereas the SOH configuration increases both dimensions to 128 to cope with the longer temporal context. LSTM gates i_t, f_t, o_t and candidate state g_t follow the standard formulation used in modern RNN surveys [35]. All models are implemented in PyTorch [36]. For one streaming update with input dimension $D = 6$, LSTM hidden size H , and fixed MLP width M , the matrix products require

$$N_{\text{MAC}}(H) = 4H(D + H) + HM + M, \quad (14)$$

where one multiply-accumulate (MAC) is counted per weight and bias additions, nonlinearities, indexing, and memory transfers are excluded. This gives 22,080 MACs for SOC Base ($H = M = 64$) and 12,124 after pruning to $H' = 45$, a 45.1 % reduction. The corresponding SOH counts are 85,120 for Base ($H = M = 128$) and 46,208 after pruning to $H' = 90$, a 45.7 % reduction. More generally, a nominal 30 % hidden-size reduction sets $H' = 0.70H$. As the quadratic recurrent term dominates for large H , the analytical MAC reduction approaches $1 - 0.70^2 = 51$ %. The linear input/MLP and fixed output terms keep the implemented models below this limit. Appendix A.6 visualises this architecture-level scaling. This relation makes the computational effect of

348 a selected hidden-size reduction predictable before deployment, while hardware
 349 measurements translate the analytical reduction into platform-specific flash, la-
 350 tency, and energy changes.

$$\begin{aligned}
 \mathbf{z}_t &= [\mathbf{x}_t, \mathbf{h}_{t-1}], \\
 \mathbf{i}_t &= \sigma(W_i \mathbf{z}_t + \mathbf{b}_i), \quad \mathbf{f}_t = \sigma(W_f \mathbf{z}_t + \mathbf{b}_f), \\
 \mathbf{o}_t &= \sigma(W_o \mathbf{z}_t + \mathbf{b}_o), \quad \mathbf{g}_t = \tanh(W_g \mathbf{z}_t + \mathbf{b}_g), \\
 \mathbf{c}_t &= \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t, \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t).
 \end{aligned} \tag{15}$$

351 Equations (14) and the recurrent update above define the common Base ar-
 352 chitecture used for both tasks. Pruning changes its hidden dimension and pa-
 353 rameter submatrices, whereas quantization changes the stored representation of
 354 selected recurrent weights. Neither transformation changes the input features,
 355 targets, or streaming state-update order.

356 Both SOC and SOH models are trained with a mean-squared-error loss be-
 357 tween the prediction and the corresponding reference at each training sample.
 358 During validation we additionally report mean absolute error (MAE), root-
 359 mean-square error (RMSE), and R^2 . The trained parameters, configuration
 360 files, and optimiser statistics are retained to enable deterministic replay of both
 361 models.

362 4.2. Training Notes

363 Training is carried out on a GPU node equipped with NVIDIA A100-class
 364 accelerators using mixed-precision (FP16) arithmetic. Both models use the
 365 AdamW optimiser with cosine-annealing warm-restart scheduling, gradient clip-
 366 ping, and mini-batch training [37, 38, 39, 40]. The SOC configuration uses
 367 a learning rate of 1.5×10^{-4} , a batch size of 512, and sequence chunks of
 368 length 2024. The corresponding SOH values are 2.0×10^{-4} , 128, and 2048.

369 Deployment operates the LSTM in stateful, step-by-step streaming mode,
 370 whereas training uses truncated backpropagation through time over these chunks.
 371 Hidden and cell states are propagated within every chunk in the same tempo-
 372 ral order as during deployment. Chunking permits randomised segment sam-

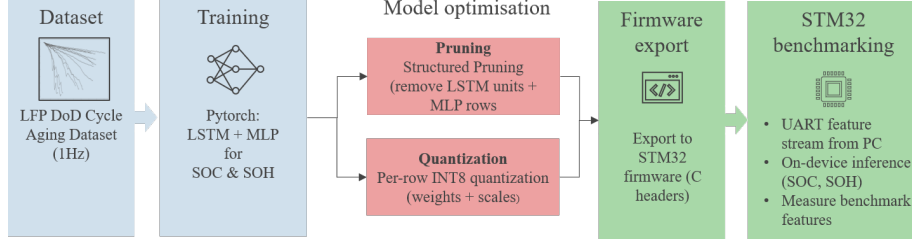


Figure 5: End-to-end pipeline from dataset to embedded benchmarking. The LFP DoE Cycle Ageing Dataset feeds PyTorch training on a GPU workstation. The resulting SOC and SOH LSTM+MLP models are pruned and quantized, exported as C headers, and deployed on an STM32H753ZI board, where flash/RAM usage, inference time, and latency are measured under streaming replay. Estimated energy per inference is derived from kernel time under a constant board-power assumption.

373 pling and tractable gradient computation without changing the recurrent update
374 equations.

375 4.3. Baseline Accuracy

376 On the workstation the SOC and SOH base models are first evaluated in
377 streaming mode on the full ageing trajectory of one representative LFP cell
378 from the LFP DoE Cycle Ageing Dataset [15], yielding baseline MAE and RMSE
379 values for both tasks. These Python replays operate on the original PyTorch
380 implementations and provide the numerical reference for the compression exper-
381 iments. The corresponding trained models are pruned, quantized, and exported
382 to STM32 firmware, where identical feature streams are replayed to measure
383 accuracy, flash and RAM usage, system latency, and on-device inference time.
384 A timing-derived energy proxy is calculated as described in Section 7. Figure 5
385 summarises this training-to-firmware benchmarking pipeline.

386 5. Pruning

387 Pruning reduces the size and computational cost of a neural network by
388 eliminating parameters or whole units that contribute little to the overall map-
389 ping, while keeping the input-output behaviour as close as possible to the dense

baseline [41, 42]. In our setting we apply structured pruning at the level of complete LSTM hidden units so that the resulting model remains well suited for efficient C implementations on the microcontroller.

5.1. Structured magnitude pruning

Structured magnitude pruning removes the lowest-saliency LSTM units until only 45 hidden channels remain in the SOC model (30% sparsity), with analogous settings for the SOH experiments. Reducing the state dimension is a prerequisite for fitting the estimators into the flash and SRAM budgets of the target STM32-based BMS and for shortening the on-device inference time, inspired by structured sparsity approaches for deep and recurrent networks [43, 44, 45]. In analogy to the group-lasso style regularisers proposed for LSTMs in [43, 44], which aggregate weights across gates via L2 norms, we define a saliency score for each hidden unit h that combines all incoming weights from the different gates:

$$s_h = \sum_{g \in \{i, f, o, g\}} \left(\|W_{ih}^{(g)}[h, :]\|_2 + \|W_{hh}^{(g)}[h, :]\|_2 \right). \quad (16)$$

Units are ranked by s_h , and the H' highest-scoring indices form the retained set $\mathcal{R}$. For each gate g , structured pruning constructs the reduced matrices

$$\begin{aligned} \widetilde{W}_{ih}^{(g)} &= W_{ih}^{(g)}[\mathcal{R}, :], \\ \widetilde{W}_{hh}^{(g)} &= W_{hh}^{(g)}[\mathcal{R}, \mathcal{R}], \\ \widetilde{W}_{MLP,1} &= W_{MLP,1}[:, \mathcal{R}]. \end{aligned} \quad (17)$$

Thus the same index set removes gate rows, recurrent columns, and the corresponding inputs of the first MLP layer. The reduced tensors remain dense. The computational saving comes from the smaller dimension rather than sparse indexing.

The final SOC deployment uses $H' = 45$ instead of 64 hidden units, while SOH uses $H' = 90$ instead of 128. Both transformations remove approximately 30 % of the recurrent channels. This is a one-shot structured operation followed

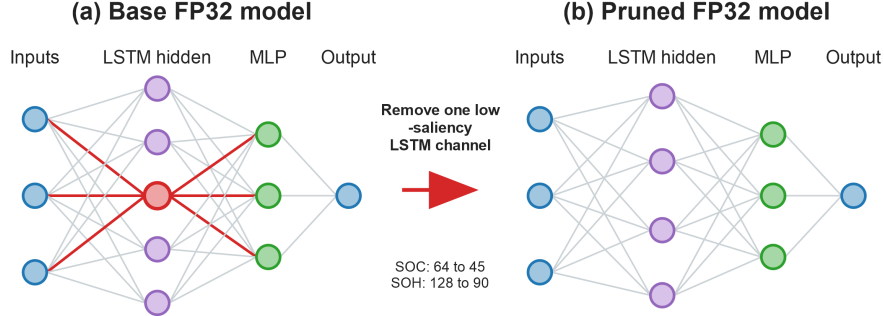


Figure 6: Structured hidden-channel pruning used in this study. Panel (a) marks one low-saliency LSTM channel and its incident connections in red. Panel (b) shows the resulting dense network after removing the corresponding gate rows, recurrent columns, and MLP input column according to Eq. (17). The final SOC and SOH hidden dimensions are shown between the panels.

by brief post-pruning fine-tuning. No iterative pruning schedule or pruning-aware training is used. The L2 aggregation gives a gate-consistent ranking that can be calculated directly from the model weights and mapped to dense C arrays. Appendix A.1 reports the resulting SOC saliency ranking and recurrent-weight distributions. These diagnostics describe the resulting weight selection but do not establish a causal regularisation effect. The criterion is selected because it combines all four gate rows for one recurrent unit and maps directly to removal of dense rows, recurrent columns, and the associated MLP input. It therefore provides a reproducible, gate-consistent ranking that translates directly into a smaller dense kernel. Gradient sensitivity would additionally use training data and backpropagation, while activation-based ranking would use a representative calibration stream and a temporal aggregation rule. A future comparison with these data-dependent alternatives can determine whether their additional information further improves the accuracy–efficiency trade-off. Appendix A.7 summarises the method properties.

428 5.2. Resulting pruned models

429 The resulting pruned SOC and SOH models still run in FP32 but reduce flash
 430 consumption and runtime considerably once compiled for STM32. Section 7
 431 summarises the corresponding latency and memory savings. These pruned vari-
 432 ants are evaluated alongside their base and quantized counterparts throughout
 433 the results section.

434 6. Quantization

435 Quantization reduces the numerical precision of selected weights while keep-
 436 ing the network topology unchanged. We apply post-training, weight-only quan-
 437 tization to the LSTM input-to-hidden and hidden-to-hidden matrices. Biases,
 438 activations, recurrent hidden and cell states, and the MLP remain FP32. The
 439 resulting firmware is therefore a mixed-precision storage implementation, not a
 440 fully integer inference path. Quantization-aware training is not used.

441 6.1. Per-row symmetric quantization

442 For row r of either recurrent matrix, the implementation computes

$$s_r = \max\left(\frac{\max_h |w_{r,h}|}{127}, \varepsilon_q\right), \quad (18)$$

$$q_{r,h} = \text{clip}\left(\text{round}\left(\frac{w_{r,h}}{s_r}\right), -127, 127\right), \quad (19)$$

$$\hat{w}_{r,h} = s_r q_{r,h}, \quad (20)$$

443 where ε_q prevents division by zero for an all-zero row. Although the INT8
 444 storage type provides 256 machine codes, the implementation deliberately leaves
 445 -128 unused and employs the symmetric 255-level set $\{-127, \dots, 127\}$. This
 446 keeps zero exactly representable and assigns equal positive and negative ranges.
 447 With the max-absolute scale in Eq. (18), finite row weights do not clip, and
 448 nearest-integer rounding gives the element-wise reconstruction bound

$$|w_{r,h} - \hat{w}_{r,h}| \leq \frac{s_r}{2}. \quad (21)$$

449 Per-row scaling gives small-magnitude rows a finer step size than one global
 450 scale, following standard uniform post-training quantization practice [46, 47].

451 6.2. Mixed-precision gate computation

452 Let r denote one gate row. The firmware evaluates

$$g_r = \sum_{d=1}^D x_d (s_{ih,r} q_{ih,r,d}) + \sum_{j=1}^H h_j (s_{hh,r} q_{hh,r,j}) + b_{fp32,r}. \quad (22)$$

453 Thus the recurrent weights are stored as INT8 codes but reconstructed into
 454 FP32 products inside the current accumulation loops. A built-in streaming
 455 comparison validates the exported gate computations against the source model
 456 and typically yields MAE differences below 10^{-4} .

457 The recurrent matrices account for approximately 80 % of the raw Base
 458 model constants in both tasks. Targeting these matrices concentrates the stor-
 459 age reduction on the dominant constants while retaining FP32 biases, activa-
 460 tions, recurrent states, and the smaller MLP path. This creates a transparent
 461 mixed-precision configuration that reduces persistent storage without changing
 462 the recurrent state representation. Quantizing activations and states could ex-
 463 tend the approach toward additional RAM and latency reductions and is consid-
 464 ered as a subsequent optimisation step. Appendix A.8 documents the precision
 465 boundary and model-constant composition.

466 6.3. Export for STM32

467 The exported headers are integrated into separate C firmware projects for
 468 the stateful SOC and SOH estimators. Each project combines the LSTM and
 469 MLP kernels with UART telemetry. A small driver library provides build files
 470 and flashing scripts so that identical measurement binaries can be rebuilt on
 471 other machines. This explicit tool chain, from training through pruning and
 472 quantization to firmware, exposes the numerical operations and records the
 473 build settings used for the resource measurements in Section 7.

474 7. Benchmark Methodology

475 This section describes how we benchmark the SOC and SOH models on both
 476 the host PC and an STM32 Nucleo-144 development board with an STM32H753ZI

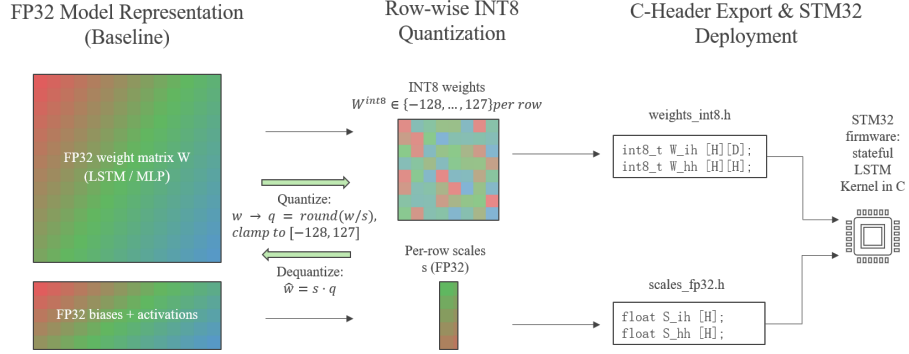


Figure 7: FP32-to-INT8 quantization and embedded deployment. The recurrent FP32 weight matrices are mapped row by row to INT8 codes and one FP32 scale per row. The INT8 storage type spans $\{-128, \dots, 127\}$, while the implemented symmetric quantizer generates $\{-127, \dots, 127\}$ and leaves -128 unused. The integer recurrent weights and scale vectors are exported as C headers and combined with FP32 biases, activations, recurrent states, and MLP operations in the STM32 kernel.

477 MCU [48, 14]. The microcontroller provides 2 MB of on-chip Flash and 1 MB of
 478 SRAM, which constitute the budget for firmware code, model parameters, and
 479 runtime buffers. Starting from the trained LSTM + MLP models, we generate
 480 three embedded variants (full-precision baseline, pruned model, and quantized
 481 model) and evaluate them on the *LFP DoE Cycle Ageing Dataset*. The bench-
 482 mark combines long-horizon streaming simulations with hardware-in-the-loop
 483 measurements and is fully scripted to ensure reproducible accuracy and resource
 484 evaluations.

485 On the host side, the complete reference ageing sequence is replayed in
 486 Python using the same streaming inference functions that are later exported
 487 to C. The SOC and SOH estimators are evaluated in continuous time without
 488 resetting their recurrent states, and error metrics (MAE, RMSE, R^2), error his-
 489 tograms, and parity plots are computed from the resulting trajectories. These
 490 simulations provide a high-precision reference for the embedded experiments
 491 and are used to assess how pruning and quantization affect the dynamics of the
 492 baseline models. To test whether deviations grow over time, the full-stream tra-

jectories are divided into ten consecutive, equal-count segments. Target MAE and P95 error are computed in every segment, together with the mean absolute deviation of each compressed variant from the Base output. Appendix A.2 reports this temporal analysis.

For the embedded benchmarks the base, pruned, and quantized SOC/SOH models are embedded into dedicated firmware projects for the STM32H753ZI. Because STM32Cube AI does not natively support stateful LSTM layers, the recurrent update equations, post-processing, and UART diagnostics are implemented manually in C. Identical stimulus streams are then injected via UART, allowing a direct comparison between host-side and on-device behaviour under the same operating conditions.

7.1. Limited software-level fault sensitivity

Two limited software-level sensitivity analyses examine the Base, Pruned, and Quantized variants under missed reports and transient input corruption. First, randomly omitted outputs at rates of 1, 5, and 10 %, as well as ten deterministic gaps of 60, 600, or 3600 samples, are replaced by the preceding prediction using a hold-last strategy.

The second analysis evaluates how a one-sample input-buffer fault propagates through the recurrent state. Each exported C model processes 8000 consecutive C07 samples at 1 Hz. Thirty independent events are distributed over this sequence after at least 2048 preceding clean samples, with ten events each for voltage, current, and temperature. At every event, the disturbed branch starts from the same recurrent state as the clean branch. Bit 22, the most significant FP32 mantissa bit, is flipped in one input value before inference for exactly one sample. The following 60 inputs are unmodified and the recurrent state is not reset.

For event e and offset $k = 0, \dots, 60$, the fault-induced deviation is $d_{e,k} = 100|\hat{y}_{e,k}^{\text{fault}} - \hat{y}_{e,k}^{\text{clean}}|$ in percentage points. Peak deviation and residual deviation after 60 s quantify the transient response independently of the model’s pre-existing target error. Accuracy is evaluated over the same 61-sample window

523 using

$$\Delta\text{MAE}_{61,e} = \frac{100}{61} \sum_{k=0}^{60} (|\hat{y}_{e,k}^{\text{fault}} - y_{e,k}| - |\hat{y}_{e,k}^{\text{clean}} - y_{e,k}|). \quad (23)$$

524 A positive value indicates that the bit flip increases target MAE, while a neg-
 525 ative value means that the disturbed prediction happens to lie closer to the
 526 target. Recovery is the first time after the peak at which $d_{e,k}$ remains below
 527 $\max(0.1d_{e,\text{peak}}, 10^{-4} \text{ pp})$ for five consecutive samples. This test represents a
 528 transient input-buffer upset in software. It does not replace hardware fault
 529 injection into weights, activations, recurrent-state memory, communication, or
 530 MCU memory.

531 7.2. STM32 Measurements

532 On the STM32 Nucleo-144 development board, dedicated SOC and SOH
 533 firmware images execute the embedded estimators while host-side scripts handle
 534 stimulus injection and UART logging. Recorded feature streams are transmit-
 535 ted to the microcontroller, which runs the stateful LSTM + MLP and returns
 536 predictions with timing diagnostics.

537 Each sample produces two time stamps. Host latency measures the delay
 538 between sending an input vector and receiving its estimate. On-device inference
 539 time is measured with the data watchpoint and trace (DWT) cycle counter
 540 around the LSTM and MLP kernels. Reported values are averages over 10,000
 541 streaming inferences. The timed interval includes instructions and memory
 542 accesses executed by these kernels but excludes UART transfer, host processing,
 543 and host-side feature construction.

544 In the SOC projects, six-feature robust scaling lies inside the timed estimator
 545 call. The SOH Base and Pruned projects apply scaling immediately before the
 546 DWT interval, whereas the Quantized SOH forward function includes it. This
 547 small affine preprocessing cost was not profiled separately and is absent from
 548 the static operation counts. All inspected benchmark projects were compiled
 549 with GCC at -O0, using the hard-float ABI and Cortex-M7 FPU floating-point
 550 unit. The measurements therefore characterise auditable, unoptimised reference
 551 kernels rather than production-optimised integer deployments.

552 7.3. Key Performance Indicators

553 For normalised target y_n and prediction $\hat{y}_n$, the signed error is expressed in
 554 percentage points as $e_n = 100(\hat{y}_n - y_n)$. The reported accuracy indicators are

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^N |e_n|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N e_n^2}, \quad (24)$$

$$\text{P95} = Q_{0.95}(\{|e_n|\}_{n=1}^N), \quad (25)$$

555 where $Q_{0.95}$ is the empirical 95th percentile. MAE and RMSE use all samples,
 556 whereas P95 identifies the absolute-error threshold below which 95 % of samples
 557 lie.

558 For each deployment we additionally report average host latency, on-device
 559 inference time, flash and SRAM usage, and an estimated energy-per-inference
 560 proxy. Tables 1 and 2 focus on these STM32 resource and timing indicators.
 561 Latency is the end-to-end UART delay, whereas inference time covers only the
 562 DWT-measured kernel interval. RAM combines statically allocated data and
 563 BSS sections with maximum observed streaming stack usage.

564 Energy was not measured separately for each model. Instead, we use

$$E_{\text{est}} = P_{\text{assumed}} t_{\text{inf}}, \quad P_{\text{assumed}} = 0.5 \text{ W}, \quad (26)$$

565 where 0.5 W is a representative average-power assumption for the complete de-
 566 velopment board. Consequently, the percentage change in E_{est} is identical to
 567 the percentage change in inference time. The proxy does not separate core com-
 568 putation, flash access, SRAM access, voltage-regulator losses, or other board
 569 consumers and cannot determine whether flash access dominates the actual en-
 570 ergy consumption. An analogous set of measurements was recorded for the SOH
 571 firmware deployments. Table 2 lists the corresponding indicators.

572 The KPIs in Tables 1 and 2 highlight the trade-offs introduced by pruning
 573 and quantization. For SOC, pruning reduces the flash footprint by roughly 41 %
 574 and inference time by about 43 %, while latency remains dominated by UART
 575 transfer. Quantization approximately halves flash usage, but per-weight scaling
 576 and integer–float conversions in the mixed-precision kernels increase on-device

Table 1: Engineering KPIs for SOC models on STM32H753ZI (full-stream averages). Energy is estimated from measured inference time under an assumed constant board power of 0.5 W.

Model	Latency [ms]	Inference [ms]	Flash [KB]	RAM [KB]	E_{est} [mJ]
Base FP32	12.70	1.40	105.32	4.93	0.70
Pruned FP32	12.20 (-3.9%)	0.80 (-42.9%)	62.27 (-40.9%)	4.03 (-18.3%)	0.40 (-42.9%)
Quant INT8	18.48 (+45.5%)	6.99 (+399%)	52.48 (-50.2%)	3.96 (-19.7%)	3.49 (+399%)

Table 2: Engineering KPIs for SOH models on STM32H753ZI (full-stream averages). Energy is estimated from measured inference time under an assumed constant board power of 0.5 W.

Model	Latency [ms]	Inference [ms]	Flash [KB]	RAM [KB]	E_{est} [mJ]
Base FP32	34.88	22.73	335.00	8.69	11.37
Pruned FP32	24.69 (-29.2%)	12.72 (-44.0%)	182.41 (-45.5%)	6.96 (-19.9%)	6.36 (-44.0%)
Quant INT8	41.23 (+18.2%)	29.21 (+28.5%)	138.00 (-58.8%)	6.70 (-22.9%)	14.60 (+28.5%)

inference time by a factor of four relative to Base. The SOH deployments show the same qualitative behaviour. Pruning saves about 46 % of flash and 44 % of inference time, whereas Quantized attains the smallest flash footprint but incurs a 29 % timing increase. Under the constant-power assumption, the estimated energy proxy follows these timing changes exactly. It is not independent evidence of board-level energy efficiency.

8. Results

The evaluation compares the full-precision Base, Pruned, and Quantized SOC and SOH estimators under the same continuous streaming replay of the LFP DoE Cycle Ageing Dataset. The following sections separate prediction accuracy, temporal behaviour, embedded resource use, and the resulting multi-objective trade-offs.

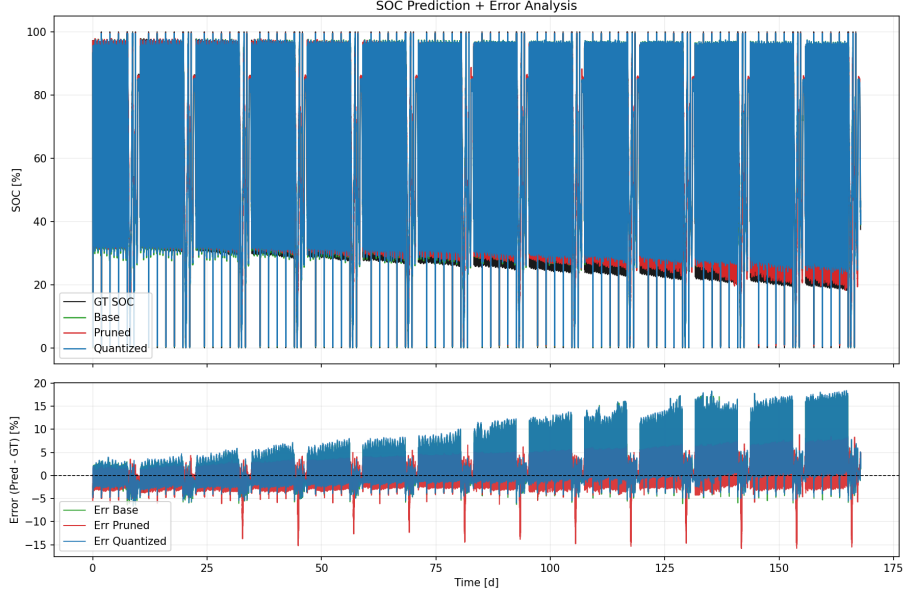


Figure 8: Streaming SOC dashboard on the full ageing trajectory. The black line represents the reference SOC, while the coloured lines show the predictions of the Base (green), Pruned (red), and Quantized (blue) models. The lower sub-panel displays the instantaneous prediction error.

8.1. Streaming Prediction Accuracy

The primary metric for any battery state estimator is its ability to track the ground truth over extended periods without divergence. Figures 8 and 11 illustrate the streaming performance of the SOC and SOH models over the complete test duration. Across the complete SOC replay, the Base, Pruned, and Quantized variants obtain MAE values of 2.685, 2.338, and 2.791 percentage points, respectively. The Quantized trace often overlaps the Base trace visually, whereas the Pruned model has the lowest aggregate target MAE despite a larger model-to-Base deviation in some parts of the sequence. These results confirm that post-training weight quantization preserves useful recurrent behaviour throughout the evaluated stream. Replays of additional operating sequences can extend this comparison to a wider range of temporal conditions.

The SOH estimator faces a different challenge: it must infer slow capacity

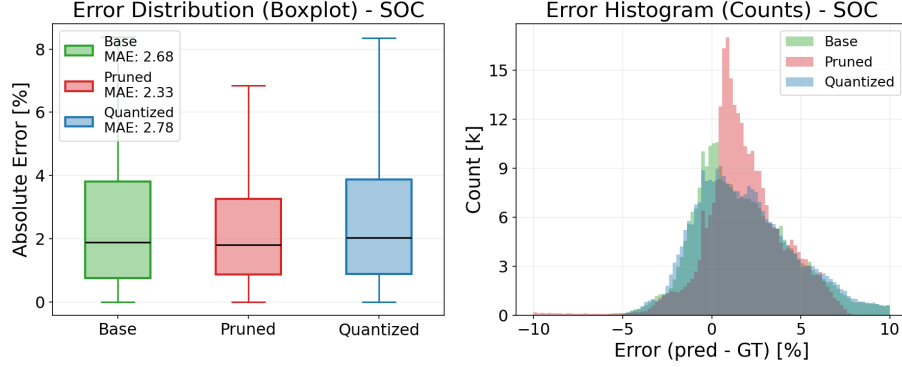


Figure 9: SOC error distributions for Base, Pruned, and Quantized. The left panel compares absolute errors and aggregate MAE. The right panel shows signed errors, $100(\hat{y} - y)$, so that distribution shifts and tails remain visible.

602 degradation from sparse check-up events and continuous monitoring. In the
 603 plotted trajectories the discrete capacity tests have been removed from the esti-
 604 mator input, while the remaining check-up blocks (OCV sweeps, HPPC pulses,
 605 and profile segments) are retained. The plateau-like sections of the reference
 606 curve therefore correspond to these diagnostic intervals. The dashboard applies
 607 the complete two-stage causal post-processing chain from Section 3.5: initial
 608 alignment, stage 1 with $\alpha_1 = 0.02$ and a symmetric limiter, followed by stage 2
 609 with $\alpha_2 = 10^{-6}$ and a downward limiter. The resulting smoothing suppresses
 610 short-term variation but introduces a substantial accuracy–delay trade-off.

611 8.2. Temporal stability and filter interaction

612 The ten-segment analysis in Appendix A.2 retains the natural temporal or-
 613 der and does not reset recurrent states between segments. SOC target error
 614 rises toward the later part of the sequence for every variant: for example, Base
 615 MAE increases from 1.116 to 4.538 percentage points and Quantized MAE from
 616 1.234 to 4.600 percentage points between the first and final tenths. However,
 617 the Quantized-to-Base mean absolute deviation changes from 0.332 to 0.271
 618 percentage points over the same interval and therefore shows no quantization-
 619 specific monotonic accumulation. The Pruned-to-Base deviation increases from

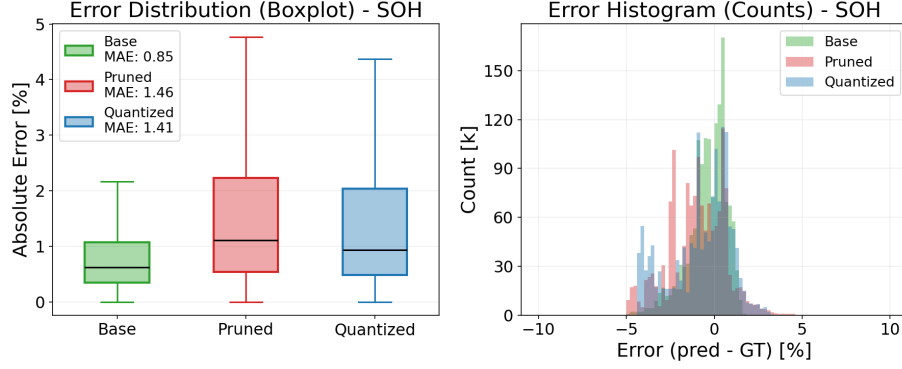


Figure 10: SOH error analysis: MAE distribution (left) and error histogram (right). The three variants retain similar central error ranges, while their tails and aggregate MAE values differ.

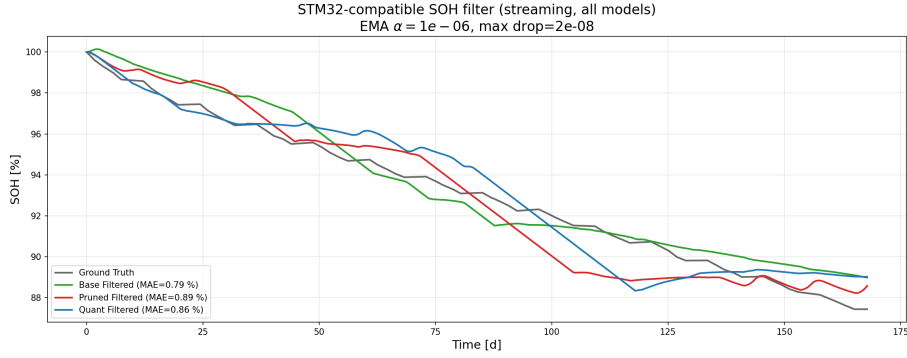


Figure 11: Streaming SOH dashboard with the complete two-stage causal post-processing chain from Section 3.5. The top panel shows the reference SOH and the outputs of the Base (green), Pruned (red), and Quantized (blue) models. The bottom panel depicts the corresponding filtered prediction errors.

620 0.924 to 2.592 percentage points even though Pruned has the lowest aggregate
621 target MAE. SOH window errors fluctuate with the ageing phase, and nei-
622 ther compressed variant exhibits monotonic model-to-Base divergence across all
623 ten segments. The analysis therefore demonstrates stable relative behaviour
624 throughout the complete evaluated replay and provides a basis for extending
625 validation to further operating sequences.

Appendix A.3 compares unfiltered, stage-1, and final two-stage SOH outputs for the three exported C models on a common input sequence. Stage 1 gives MAE values of 1.677, 1.644, and 1.926 percentage points for Base, Pruned, and Quantized, respectively. Applying stage 2 to the stage-1 output gives final MAE values of 1.476, 1.695, and 1.028 percentage points. The identical input and processing order reveal how filtering changes both the absolute error and the ranking of the model variants. Combining raw and filtered accuracy with the filter response time therefore provides a more complete assessment of the estimator dynamics. The final EMA has an EMA-only 90 % response time of 26.65 days at 1 Hz.

8.3. Error Distribution Analysis

To quantify the reliability of the estimators, we analyse the statistical distribution of the prediction errors. Figures 9 and 10 present the Mean Absolute Error (MAE) boxplots and the error histograms for both tasks. The SOC error distribution is narrow and approximately centred around zero. The evaluated Pruned model attains the smallest aggregate MAE, while the Quantized model remains close to the Base result with a smaller recurrent-weight footprint. Appendix A.1 shows that the 19 removed SOC units occupy the lowest positions of the defined L2 saliency ranking and compares the central recurrent-weight distributions before and after pruning. This confirms that the implemented criterion removes the intended low-saliency channels and identifies Pruned as the most accurate variant for the evaluated model and data partition. Repeated training runs can determine the statistical consistency of this advantage without attributing it to a specific regularisation mechanism. For SOH, the error distribution is broader, which is inherent to the difficulty of estimating capacity fade from operational data. However, the histograms show that the bulk of the errors are still contained within a useful tolerance band (typically $\pm 2\%$), making the embedded estimator a viable tool for onboard diagnostics.

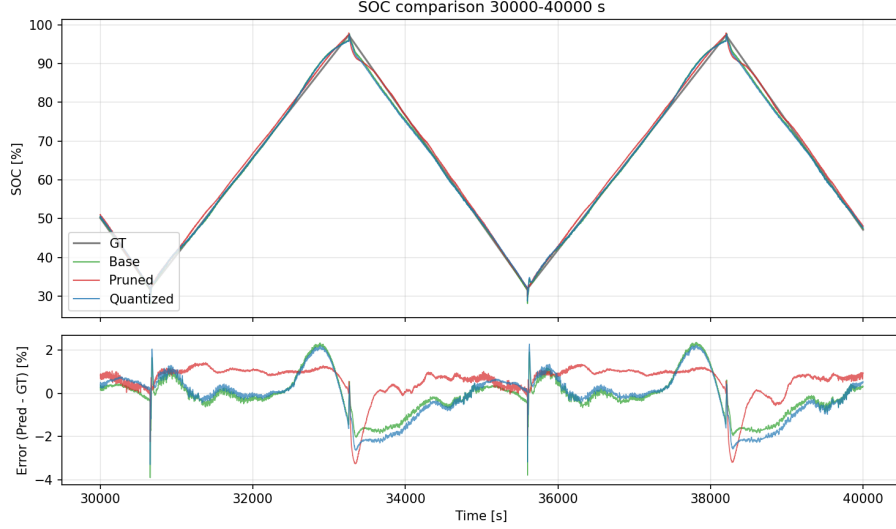


Figure 12: Detailed transient analysis of SOC streaming replay (pulse segment). Response to dynamic load pulses (30k–40ks).

8.4. Transient Response and Zoomed Analysis

Global metrics often hide local failures. To inspect the transient behaviour, we zoom into a high-dynamic pulse discharge phase (Fig. 12) and into a representative check-up sequence (Fig. 13).

In the pulse region (Fig. 12), the models must react to rapid current changes. The quantized model follows the ground truth transients with high fidelity, capturing the voltage recovery phases accurately. The pruned model, having fewer parameters to model the complex electrochemical relaxation, tends to smooth out these sharp features slightly. During the check-up phase (Fig. 13), where the cell undergoes full charge/discharge cycles, all models converge well, demonstrating stability over long integration horizons.

8.5. Resource Efficiency and Latency

Figures 14 and 15 summarise the flash memory, RAM usage, parameter counts, and latency distributions for the SOC and SOH deployments on the STM32H7 microcontroller. For all deployments the model complexity is first

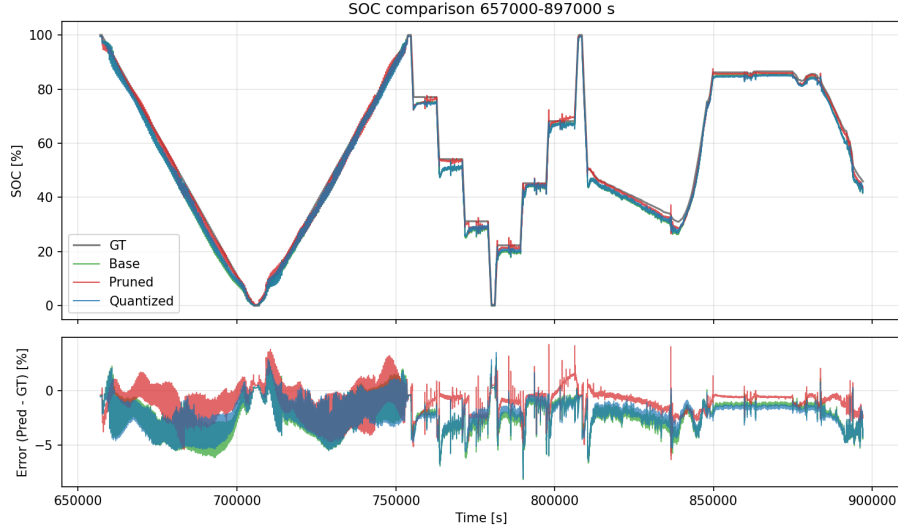


Figure 13: Detailed transient analysis of SOC streaming replay (check-up segment). Behaviour during a capacity check-up (657k–897k s).

669 characterised on the workstation side by counting parameters in each variant
 670 (base, pruned, quantized). Assuming four bytes per weight for floating point rep-
 671 resentations and one byte per weight for INT8, this directly yields an estimate of
 672 the flash footprint attributable to the neural network weights alone. The second
 673 stage of the analysis uses the linker map files of the STM32CubeIDE projects
 674 to determine the actual binary size in flash, which also includes code, constant
 675 tables, and run-time support libraries. This explains why the measured flash
 676 usage in Figure 14 is systematically higher than the idealised "weights only"
 677 estimate, and why the gap is larger for the more complex SOH firmware that
 678 contains additional diagnostic and filtering logic. The RAM usage shown in the
 679 lower right panel of Figure 14 is obtained from on-device measurements of the
 680 SOH and SOC firmware and applied analogously to all STM32 deployments.
 681 At boot time the firmware reports the size of the statically allocated data and
 682 BSS sections, which correspond to persistent variables and network parameters
 683 stored in SRAM. During each benchmark inference the minimum value of the
 684 stack pointer is tracked, so that the maximum additional stack consumption of

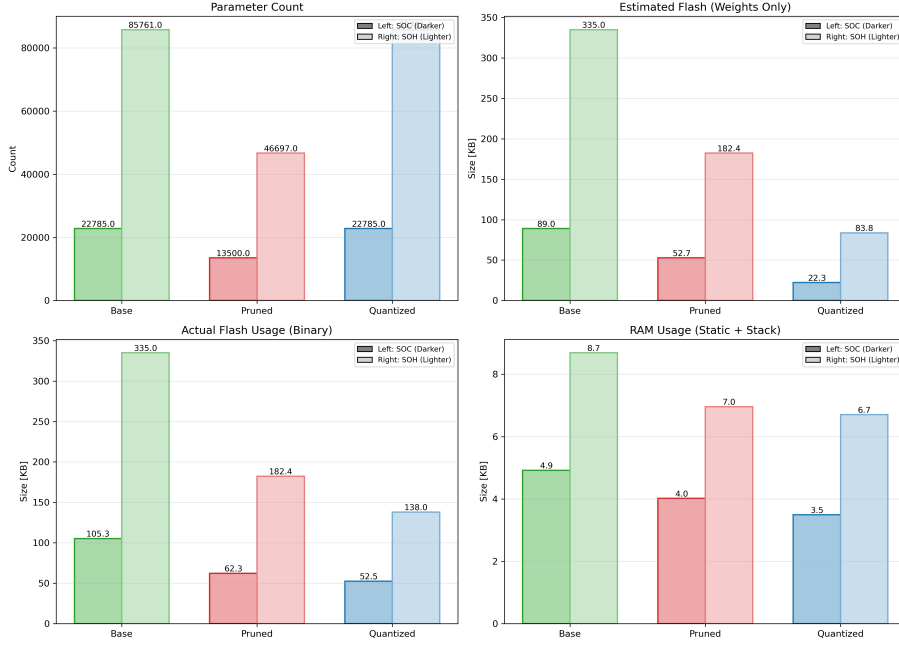


Figure 14: Resource landscape for SOC and SOH models. Comparison of parameter count, estimated Flash, actual Flash, and RAM usage. Dark bars denote SOC, light bars denote SOH.

685 the call chain can be recovered. The total RAM requirement reported in the fig-
 686 ure is the sum of static data and worst-case stack usage, i.e. an upper bound on
 687 the memory needed to execute one inference including the LSTM state updates
 688 and UART handling. This procedure captures the fact that pruning reduces
 689 both the number of parameters and the recurrent state dimension, whereas the
 690 weight-only quantized implementation primarily reduces recurrent-matrix stor-
 691 age. Activations, hidden and cell states, biases, and the MLP remain FP32.
 692 Figure 14 shows that Quantized produces the smallest flash footprint, while
 693 Pruned reduces both the parameter count and recurrent-state dimension. The
 694 measured RAM and execution-time results must nevertheless be considered sep-
 695 arately because retaining FP32 activations and changing the kernel arithmetic
 696 prevents storage reduction from translating automatically into lower latency.
 697 Figure 15 reports the corresponding host-side latency distributions. The latency

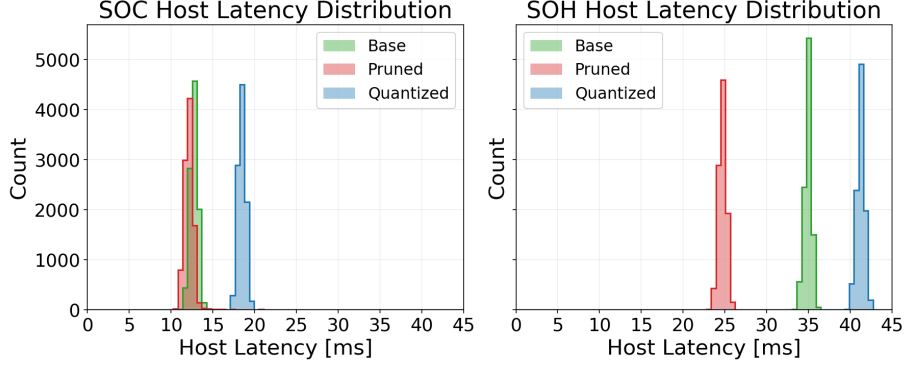


Figure 15: Latency distributions for SOC and SOH models on the STM32H7. Histograms are computed from repeated on-device measurements.

distributions in Figure 15 reveal a distinct hierarchy. SOC inference requires approximately 0.8 ms to 7 ms, whereas the SOH variants require 12 ms to 30 ms per update. The weight-quantized SOC deployment is slowest because the handwritten mixed-precision kernel applies FP32 row scales during the recurrent matrix products. When UART framing and host processing are included, end-to-end latencies remain below roughly 20 ms for SOC and 50 ms for SOH.

8.6. Composite trade-off summary

To make the multi-objective trade-offs easier to interpret at a glance, we report a composite utility score U that aggregates accuracy and embedded resource indicators into a single, dimensionless index. For task-specific non-negative weights $\mathbf{w} = (w_A, w_F, w_R, w_E)$ with $\sum_j w_j = 1$, model variant v is scored as

$$U_v(\mathbf{w}) = w_A \frac{\text{MAE}_v}{\text{MAE}_{\text{Base}}} + w_F \frac{F_v}{F_{\text{Base}}} + w_R \frac{R_v}{R_{\text{Base}}} + w_E \frac{E_{\text{est},v}}{E_{\text{est,Base}}}, \quad (27)$$

where MAE is the streaming mean absolute error, F is measured flash, R is measured total RAM (static plus peak stack), and E_{est} is the timing-derived energy proxy. The main comparison uses equal weights, $w_A = w_F = w_R = w_E = 0.25$. In this case, U is the arithmetic mean of the four normalized objectives. By construction $U_{\text{Base}} = 1$ for every valid weight vector. Lower

Table 3: At-a-glance effect of pruning and quantization relative to the corresponding Base model. Values are reported as SOC/SOH. ΔMAE is reported in percentage points (pp) of full scale, where a negative value indicates improved accuracy. Flash, RAM, and estimated-energy changes are relative deltas versus Base. The composite score U uses equal weights in (27) (lower is better).

Variant	ΔMAE [pp]	ΔFlash [%]	ΔRAM [%]	ΔE_{est} [%]	U
Pruned	-0.35/+0.61	-41/-46	-18/-20	-43/-44	0.71/0.91
Quantized	+0.10/+0.56	-50/-59	-20/-23	+399/+29	1.83/1.03

values are preferred. Because E_{est} is proportional to inference time under the fixed-power assumption, the energy term represents timing priority rather than an independent power measurement.

Sensitivity is evaluated over all 1,771 non-negative combinations on a 0.05 grid that sum to one. Appendix A.4 also varies one highlighted objective from 25 to 85 %, sharing the remaining weight equally among the other three. Pruned is top-ranked for 94.64 % of SOC combinations and 53.36 % of SOH combinations. Quantized wins 5.36 % and 17.79 %, respectively, while Base wins 0 % for SOC and 28.85 % for SOH. The ranking is therefore robust for SOC Pruned but application-dependent for SOH: Base becomes preferable when accuracy dominates, whereas Quantized becomes preferable when flash receives high weight. Table 3 summarises the resulting changes for pruning and quantization. Values are reported as SOC/SOH.

8.7. Limited software fault sensitivity

In the missing-update analysis, holding the preceding output for up to 10 % randomly omitted reports changes aggregate MAE by less than 0.001 percentage points. Ten gaps of 3600 samples increase SOC MAE by approximately 0.05 percentage points and have negligible influence on the slowly varying SOH output.

Figure A.20 shows the recurrent response to the one-sample input-buffer faults. Across the SOC variants, median peak deviation ranges from 3.29 to 3.82 percentage points and P95 ranges from 16.18 to 23.81 percentage points.

737 All 30 events per SOC model meet the recovery criterion within 60 s, with
738 median recovery times of 10.0–12.5 s. Quantized follows Base closely, whereas
739 Pruned has a lower median peak but the largest P95 peak. For SOH, median
740 peak deviation ranges from 0.83 to 1.59 percentage points and P95 from 3.37
741 to 5.85 percentage points. The share recovered within 60 s is 93.3 % for Base,
742 90.0 % for Pruned, and 86.7 % for Quantized. Among the recovered SOH events,
743 median recovery is 9, 7, and 5 s, respectively.

744 The target-based measure in Eq. 23 complements these disturbed-clean devi-
745 ations. For SOC, median ΔMAE_{61} ranges from 0.046 to 0.079 percentage points
746 and P95 from 1.42 to 1.59 percentage points. For SOH, median changes lie be-
747 tween -0.0009 and 0.0012 percentage points and P95 between 0.053 and 0.122
748 percentage points. The small negative SOH values occur when a disturbance
749 moves an already imperfect prediction closer to the target and are not evidence
750 of improved fault robustness. Table A.4 gives the model-specific values. The
751 test therefore identifies different transient response tails across the compressed
752 variants without indicating persistent SOC divergence or non-finite outputs. Its
753 restricted fault location and event count do not support a hardware-safety claim.

754 9. Discussion

755 The experiments show that pruning and quantization affect the SOC and
756 SOH estimators differently. For SOC, the evaluated Pruned model obtains a
757 0.35-percentage-point lower MAE than Base, while Quantized remains 0.11
758 percentage points above Base. Together with the L2 saliency and weight-
759 distribution diagnostics, this identifies structured pruning as the more balanced
760 transformation for the evaluated model and split. The experimental design uses
761 one training run and a predefined test split for each task because the focus
762 is the embedded comparison of compression routes. Repeated-seed validation,
763 cross-validation, and matched unpruned fine-tuning can extend the statistical
764 assessment of the observed advantage. The result is therefore reported without
765 assigning it to a specific regularisation mechanism.

766 For SOH the picture is more nuanced. Targets evolve slowly and are an-
 767 chored by periodic capacity check-ups, while the displayed online estimates
 768 undergo two strong filtering stages. The filter-stage comparison shows that
 769 post-processing can alter not only absolute MAE but also the ranking between
 770 Base, Pruned, and Quantized. Compression quality should therefore be assessed
 771 on both raw and filtered outputs, and filter selection should consider response
 772 delay rather than MAE alone.

773 From an engineering perspective the three deployment variants represent dis-
 774 tinct operating points. Base FP32 offers the simplest tool flow but requires the
 775 largest flash footprint. Pruning reduces the recurrent state dimension and the
 776 corresponding MLP input width, shrinking SOC flash from 105.32 to 62.27 kB
 777 and inference time from 1.40 to 0.80 ms. For SOH the corresponding changes
 778 are 335.00 to 182.41 kB and 22.73 to 12.72 ms. Within this unchanged FP32
 779 kernel family, static MAC ratios and measured time ratios agree closely. SOC
 780 Pruned retains 54.9 % of Base MACs and 57.2 % of its time, while SOH retains
 781 54.3 % and 56.0 %, respectively.

782 Weight-only quantization gives the smallest flash footprints, 52.48 kB for
 783 SOC and 138.00 kB for SOH, but changes the operation types without providing
 784 a fully integer execution path. Base and Quantized retain the same 22,080 SOC
 785 and 85,120 SOH model MACs. In the current C expressions, each recurrent
 786 INT8 weight is cast to FP32 and multiplied by its row scale inside the innermost
 787 accumulation, adding 17,920 source-level FP32 scale multiplications for SOC
 788 and 68,608 for SOH. Treating one MAC and one additional multiplication as
 789 one operation each gives a static Quantized/Base ratio of approximately 1.81
 790 for both tasks, whereas the measured time ratios are 4.99 for SOC and 1.29 for
 791 SOH.

792 The discrepancy is expected because the count omits conversion, indexing,
 793 loop, load/store, nonlinear-function, and cache/Flash effects and because the
 794 -00 kernel uses no optimised integer dot-product path. For the larger SOH
 795 model, reduced recurrent-weight traffic may offset more of this arithmetic over-
 796 head than for SOC. Memory bandwidth was not measured, so this remains

an implementation-based interpretation rather than a causal attribution. Appendix A.9 places the static counts beside the measured DWT times.

The present kernel prioritises a transparent implementation of the mixed-precision path. Potential improvements include accumulating each row before applying its shared scale, fusing scale and bias work, using DSP-optimised or integer dot-product kernels, and quantizing activations and recurrent states to enable a complete integer path. Operation-level cycle attribution and memory-bandwidth profiling are reserved for future implementation studies. The present results therefore establish total kernel timing for the tested firmware but not the fraction caused by conversion, arithmetic, or memory access.

The distinction between system latency, inference time, and estimated energy is particularly relevant for resource-constrained BMS. The UART-driven tests report end-to-end latency, while on-device inference spans 0.80–6.99 ms for SOC and 12.72–29.21 ms for SOH. The reported E_{est} values are obtained by multiplying these times by an assumed constant 0.5 W. They do not isolate CPU, flash, SRAM, UART, or regulator contributions. Consequently, the results establish timing and memory trade-offs on the STM32H753ZI, but not a component-level energy budget or a conclusion that flash access dominates consumption.

The utility score makes the competing effects of compression directly comparable by expressing accuracy, flash, RAM, and the timing-derived energy proxy relative to the same Base model. A value below one indicates that the combined advantages outweigh the disadvantages for the selected priorities, while a value above one indicates an overall deterioration. With equal weights, $U = 0.71$ identifies SOC Pruned as a balanced improvement, whereas $U = 1.83$ shows that the flash reduction of SOC Quantized does not compensate for its timing overhead. The SOH scores remain close to one, indicating a genuine trade-off without one consistently dominant variant. The weight-grid analysis adds the important information of how stable these conclusions are when application priorities change. SOC Pruned remains preferred across most weight combinations, while the preferred SOH model depends more strongly on whether accuracy or

flash receives greater emphasis. The utility analysis therefore supports a transparent selection of the model variant and distinguishes robust recommendations from priority-dependent decisions.

The implementation-level storage reduction follows directly from the architecture. Removing the same number of hidden channels or storing the same recurrent matrices as INT8 therefore produces predictable parameter-count changes for a fixed model topology. The preservation of estimation accuracy depends more strongly on the cell chemistry, operating conditions, reference construction, and training data. This study quantifies the trade-off for JGC LFP cells at 25 °C. Existing embedded studies across different cell chemistries demonstrate that compact data-driven estimators are broadly feasible [11, 24, 49]. Validation with additional cell chemistries is needed to determine how closely their attainable MAE and preferred compression level follow the trends observed here.

The present study examines two post-training compression routes. The first is one-shot structured pruning with short fine-tuning, and the second is recurrent-weight quantization of the original FP32 model. Quantization-aware training and pruning-aware or iterative training can in principle recover accuracy by adapting parameters to the imposed representation [47, 50]. Evaluating these strategies requires a separate training study and is reserved for future work. The current comparison therefore addresses deployable post-training transformations, not the best achievable result of jointly trained compression.

The analytical MAC relation provides a platform-independent estimate of how structured pruning changes model complexity. The measured flash and timing ratios complement this estimate for the evaluated STM32 implementation and build configuration. Their transfer to other embedded platforms depends on clock rate, floating-point and signal-processing support, compiler optimisation, cache, and memory organisation. Benchmarks on additional controller architectures can therefore quantify how these implementation characteristics translate the same model reduction into practical resource savings.

Preparing the derivative features on the host separates feature construction

859 from neural-network inference and allows the measured timing differences to be
 860 attributed directly to pruning and quantization. The long-horizon analysis eval-
 861 uates continuous inference trajectories, while the transient input-buffer analysis
 862 perturbs a measurement before inference and follows its effect through the re-
 863 current state. The latter distinguishes immediate peak deviation, recovery, and
 864 target-MAE change. It indicates similar SOC behaviour for Base and Quan-
 865 tized, a larger upper-tail SOC response for Pruned, and a larger share of SOH
 866 events without confirmed 60-s recovery for Quantized. A complete embedded
 867 front-end and safety assessment can build on these results by additionally cov-
 868 ering variable sampling, derivative filtering, recurrent-state and weight memory,
 869 activations, communication, and MCU memory.

870 **10. Conclusion and Outlook**

871 This study set out to quantify how two transparent compression routes
 872 change the accuracy and embedded cost of stateful SOC and SOH estimators un-
 873 der a common STM32 benchmark. Its contribution is an auditable comparison
 874 from trained LSTM + MLP models to continuous C inference with consistent ac-
 875 curacy, memory, and timing indicators. Estimated energy per inference remains
 876 a constant-power timing proxy and is not an independent power measurement.

877 For SOC, Pruned reduces flash by 40.9 %, RAM by approximately 18 %, and
 878 inference time by 42.9 %, while its MAE is 0.35 percentage points lower than
 879 Base for the evaluated model and split. Quantized reduces flash by 50.2 % and
 880 RAM by approximately 20 %, but increases inference time by 399 % and MAE
 881 by 0.11 percentage points in the mixed-precision software kernel. For SOH,
 882 Pruned reduces flash by 45.5 %, RAM by approximately 20 %, and inference
 883 time by 44.0 %, with a 0.61-percentage-point MAE increase. Quantized reduces
 884 flash by 58.8 % and RAM by approximately 23 %, while increasing inference
 885 time by 28.5 % and MAE by 0.56 percentage points. Under the fixed-power
 886 assumption, the energy proxy changes by the same percentages as inference
 887 time.

888 These results identify structured pruning as the more balanced operating
889 point for the tested firmware, whereas recurrent-weight quantization is attrac-
890 tive primarily when flash capacity dominates and its dequantization overhead
891 is acceptable. Ten consecutive replay segments show no quantization-specific
892 monotonic deviation from Base. After transient input-buffer bit flips, every
893 SOC variant recovers in all 30 events within 60 s, while 6.7–13.3 % of the SOH
894 events do not meet the defined recovery criterion within this horizon. The
895 SOH filter analysis further shows that compression must be assessed together
896 with post-processing delay. The evaluated LFP trajectories, model configura-
897 tions, STM32H753ZI implementation, and compiler settings establish the trade-
898 off under one complete deployment scenario. Validation across additional cell
899 chemistries, embedded platforms, model initialisations, and internal fault classes
900 can determine how broadly the observed trends transfer.

901 The first priority for future work is to optimise and profile the mixed-
902 precision kernel through row-scale hoisting, operator fusion, DSP-oriented dot
903 products, and activation/state quantization, while measuring per-model board
904 power with external instrumentation. A second priority is external validation
905 across random seeds, temperatures, cell chemistries, and embedded platforms,
906 including direct comparison with quantization-aware and pruning-aware train-
907 ing. Finally, internal weight, activation, recurrent-state, sensor, communication,
908 and MCU fault injection is required before safety-related claims can be made.
909 Joint pruning and quantization remains a promising route for combining a re-
910 duced state dimension with compact recurrent storage.

911 **Appendix A. Supplementary compression and robustness analyses**

912 This appendix presents supporting analyses of pruning structure, tempo-
913 ral stability, filter interaction, utility sensitivity, transient input faults, model-
914 complexity scaling, mixed precision, and kernel operation counts.

915 *Appendix A.1. Structured-pruning diagnostics*

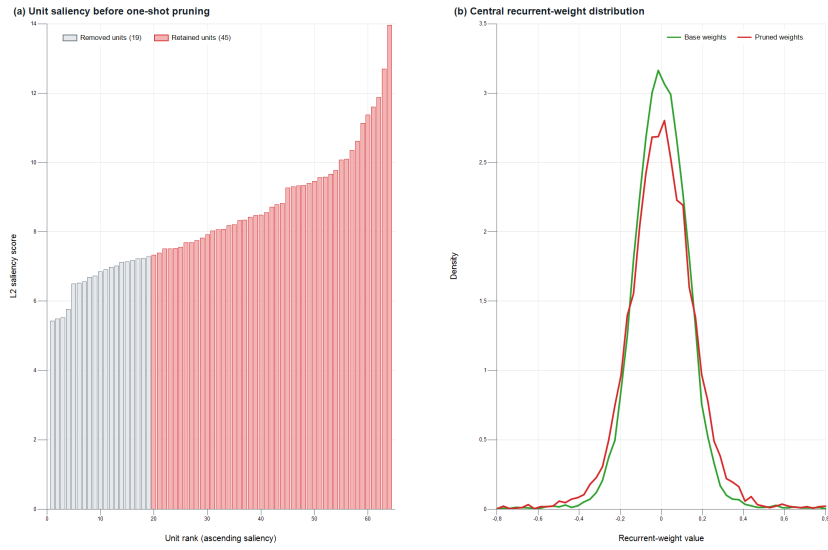


Figure A.16: Diagnostics for the one-shot SOC pruning operation. (a) L2 saliency from Eq. 16, sorted in ascending order. The 19 lowest-ranked units are removed and 45 units are retained. (b) Density of the central recurrent-weight range for the Base and Pruned models. The analysis confirms application of the specified ranking criterion and makes the resulting weight selection transparent. Repeated training runs can extend this structural evidence toward a statistical assessment of the observed MAE difference.

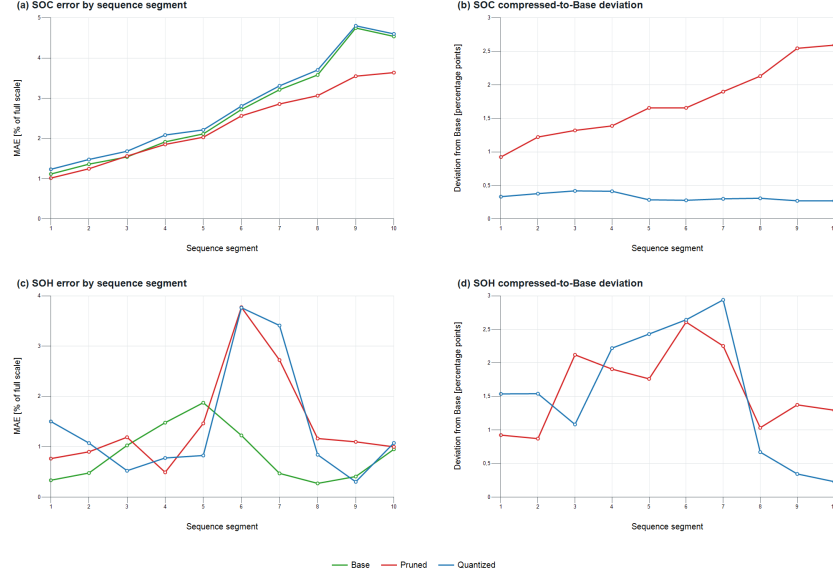


Figure A.17: Temporal error and model-to-Base deviation over ten consecutive, equal-count segments of the full-stream replay. Recurrent states are not reset at segment boundaries. Panels (a) and (c) show target MAE for SOC and SOH. Panels (b) and (d) show the mean absolute deviation of Pruned and Quantized from Base. The absence of monotonic late-sequence divergence demonstrates stable relative behaviour across the complete evaluated replay. Further sequences can extend the tested operating horizon.

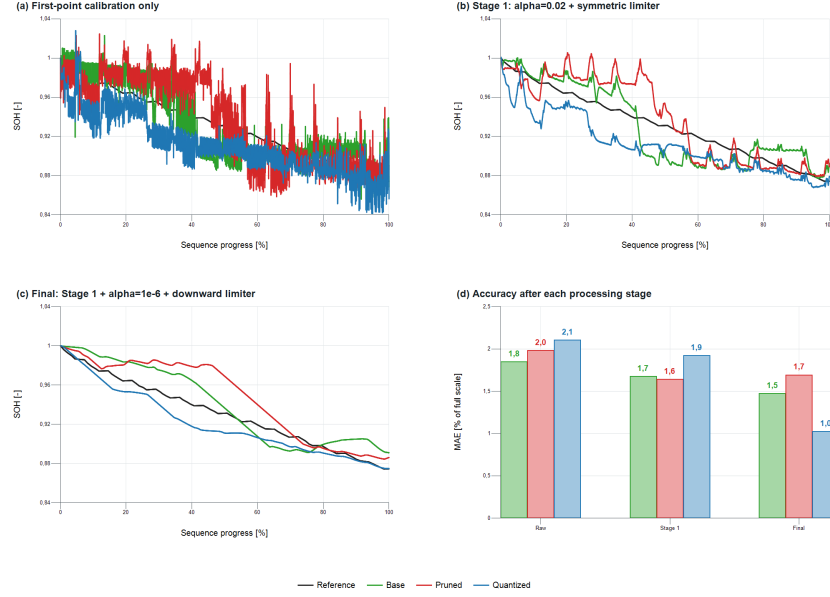


Figure A.18: Comparison of SOH compression and filtering for the Base, Pruned, and Quantized C models on a common input sequence. (a) Outputs after first-point alignment but before temporal filtering. (b) Stage 1 with $\alpha_1 = 0.02$ and the symmetric limiter. (c) Final sequential output after applying stage 2 with $\alpha_2 = 10^{-6}$ and the downward limiter to stage 1. (d) MAE after each processing stage. At 1 Hz, the EMA-only 90 % response times are approximately 114 s and 26.65 days, respectively. The nonlinear limiters have no single cutoff frequency. All three variants use the same input and processing order to isolate the interaction between compression and filtering.



Figure A.19: Sensitivity of the utility ranking to application priorities. Panels (a) and (b) vary the highlighted objective from 25 to 85 % and divide the remaining weight equally among the other three objectives. Each square identifies the lowest-utility model. At 25 %, all four objectives have equal weight. Panel (c) gives the winning share across all 1,771 non-negative 0.05-spaced weight combinations that sum to one. The energy objective uses the timing-derived E_{est} proxy and is therefore not an independent power measurement.

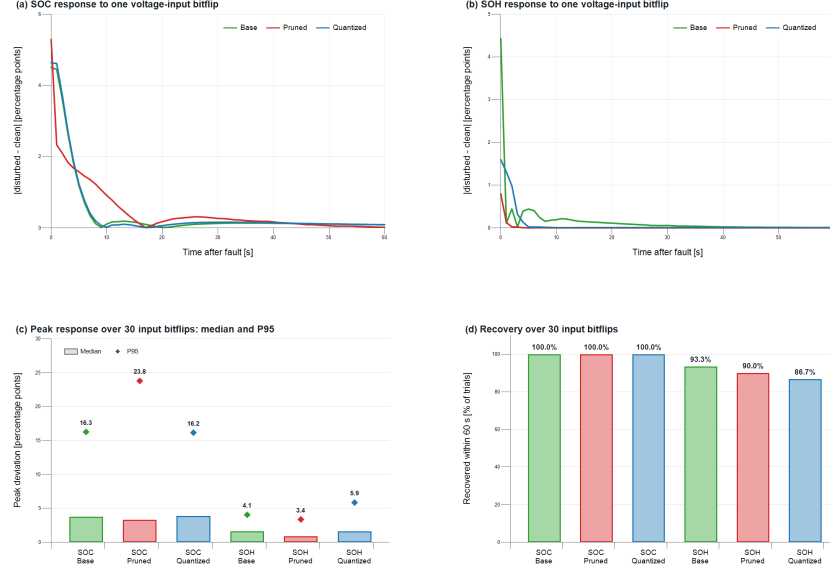


Figure A.20: Stateful response to transient software input-buffer corruption. Thirty independent events per task and model flip the most significant FP32 mantissa bit in voltage, current, or temperature for one 1-Hz sample, with ten events per input signal. The clean and disturbed branches start from the same recurrent state and then receive 60 identical clean samples without a reset. (a) and (b) show the absolute disturbed-clean deviation for one representative voltage event. (c) summarises the median peak and P95 peak across all 30 events. (d) gives the fraction that meets the sustained recovery criterion within 60 s. The figure characterises propagation through the recurrent state, while Table A.4 reports the corresponding change in target MAE.

Table A.4: Change in target MAE over the 61-sample fault and recovery window. Values are percentage points and summarise 30 independent input-bitflip events per task and model. The change is defined by Eq. 23. Negative values indicate that the disturbed output happens to lie closer to the target in part of the window.

Task	Model	Median ΔMAE_{61} [pp]	P95 ΔMAE_{61} [pp]
SOC	Base	0.0740	1.4235
SOC	Pruned	0.0464	1.5867
SOC	Quantized	0.0785	1.4314
SOH	Base	-0.0009	0.1224
SOH	Pruned	-0.00004	0.0533
SOH	Quantized	0.0012	0.1055

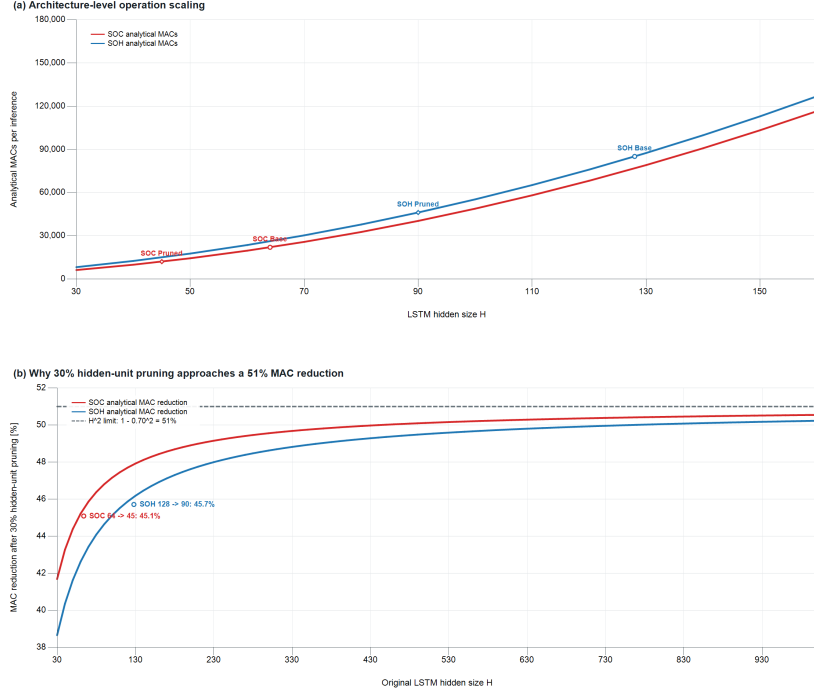


Figure A.21: Architecture-level operation scaling from Eq. 14. (a) Analytical MACs per streaming inference over LSTM hidden size, with markers for the implemented SOC and SOH Base and Pruned dimensions. (b) MAC reduction for a nominal 30 % hidden-size reduction, $H' = 0.70H$. As the quadratic recurrent term dominates, the reduction approaches $1 - 0.70^2 = 51\%$. Linear input/MLP and fixed output terms keep finite models below this limit. The relation enables the computational benefit of pruning to be estimated across model sizes before platform-specific measurements are performed.

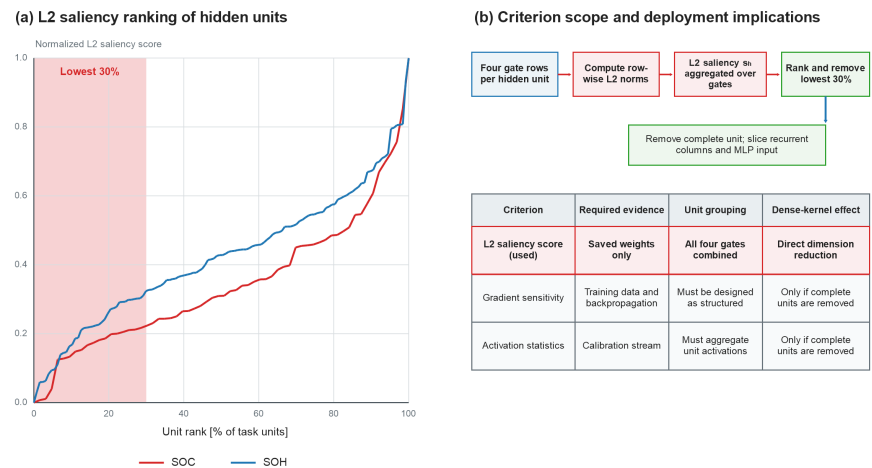


Figure A.22: Scope of the structured pruning criterion. (a) Normalised L2 saliency ranks for the SOC and SOH Base hidden units. The lowest 30 % define the task-specific removal sets. (b) The implemented path aggregates row-wise L2 norms across all four gates and removes complete recurrent units together with associated recurrent columns and MLP inputs. The comparison makes the reproducibility and dense-kernel benefit of the L2 criterion explicit and provides a reference for future comparisons with gradient- and activation-based rankings.

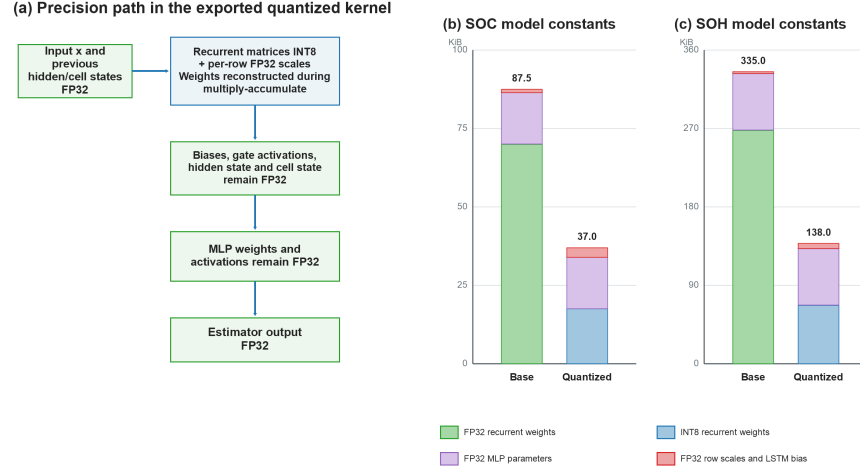


Figure A.23: Verified precision path and raw model-constant composition. (a) Only recurrent input-to-hidden and hidden-to-hidden matrices are stored as INT8. Per-row scales, biases, gate activations, hidden/cell states, the MLP, and the estimator output remain FP32. (b,c) Analytical model constants for SOC and SOH Base and Quantized variants. The comparison shows how concentrating quantization on the dominant recurrent matrices reduces persistent model storage and where activation and state quantization could provide additional runtime savings.

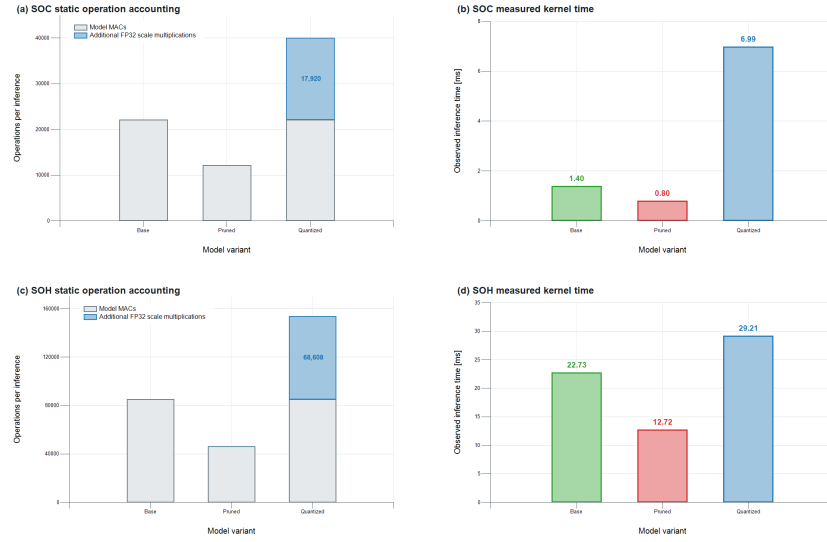


Figure A.24: Static source-level operation accounting beside measured STM32H753ZI kernel time. Panels (a,c) show model MACs and the additional FP32 row-scale multiplications implied by the current per-weight C expressions. Panels (b,d) show DWT measurements averaged over 10,000 streaming inferences. Their comparison reveals where analytical operation counts explain the timing trend and where conversion, indexing, loop, load/store, nonlinear-function, compiler, and memory-system effects create additional optimisation opportunities.

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938 **CRedit authorship contribution statement**

939 Florian Rzepka: Conceptualization, methodology, software, validation, for-
940 mal analysis, investigation, data curation, writing–original draft preparation,
941 writing–review and editing, visualization.

942 Julia Kowal: Resources, supervision, project administration, funding acqui-
943 sition, writing–review and editing.

944 **Declaration of competing interest**

945 The authors declare that they have no known competing financial interests or
946 personal relationships that could have appeared to influence the work reported
947 in this paper.

948 **Declaration of generative AI and AI-assisted technologies in the manuscript**
949 **preparation process**

950 During the preparation of this work the author(s) used ChatGPT, DeepL,
951 and Grammarly in order to improve the linguistic quality and style of the
952 manuscript. After using these tools, the author(s) reviewed and edited the
953 content as needed and take(s) full responsibility for the content of the published
954 article.

955 **Data availability**

956 The LFP DoE Cycle Ageing Dataset used in this work can be obtained from
957 the DOI repository in [15]. The landing page provides licensing information and
958 download instructions.

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